Announcements.

--Thoughts on how to study

--Final exam dates are out (May 1st or 4th)

--Recording the lecture for next Tuesday.

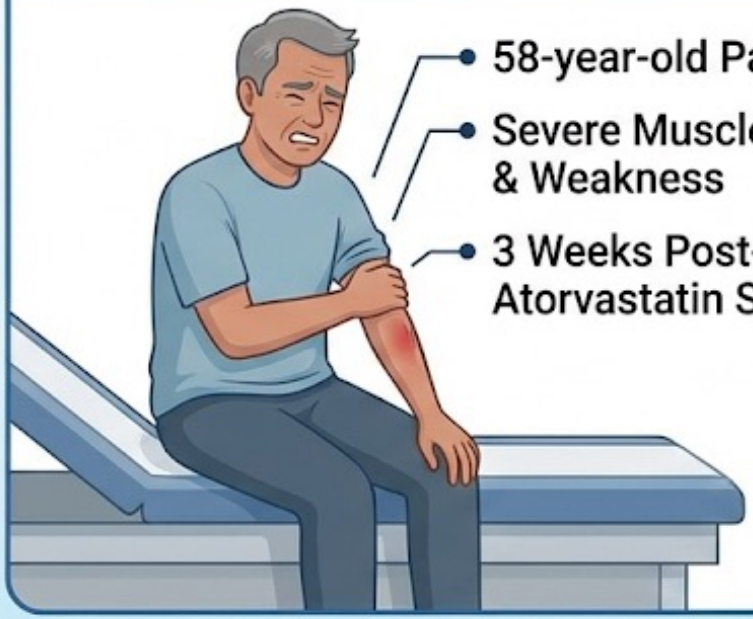
--I had slide decks from the textbook maker.

--Quick comment on procedure for taking the exam next Tuesday.

- --Bringing ID.
- --4 function calculator.
- --Room assignments according to pod.
- --Literally cutting corners.

CLINICAL CASE STUDY: THE STATIN MYSTERY

PATIENT PRESENTATION



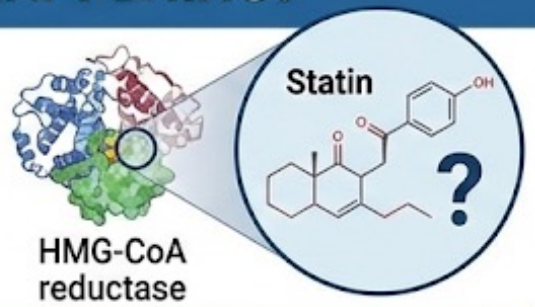
- 58-year-old Patient
- Severe Muscle Pain & Weakness
- 3 Weeks Post-Atorvastatin Start

KEY LAB FINDING

CK (Creatine Kinase) Levels:
ELEVATED 10-FOLD

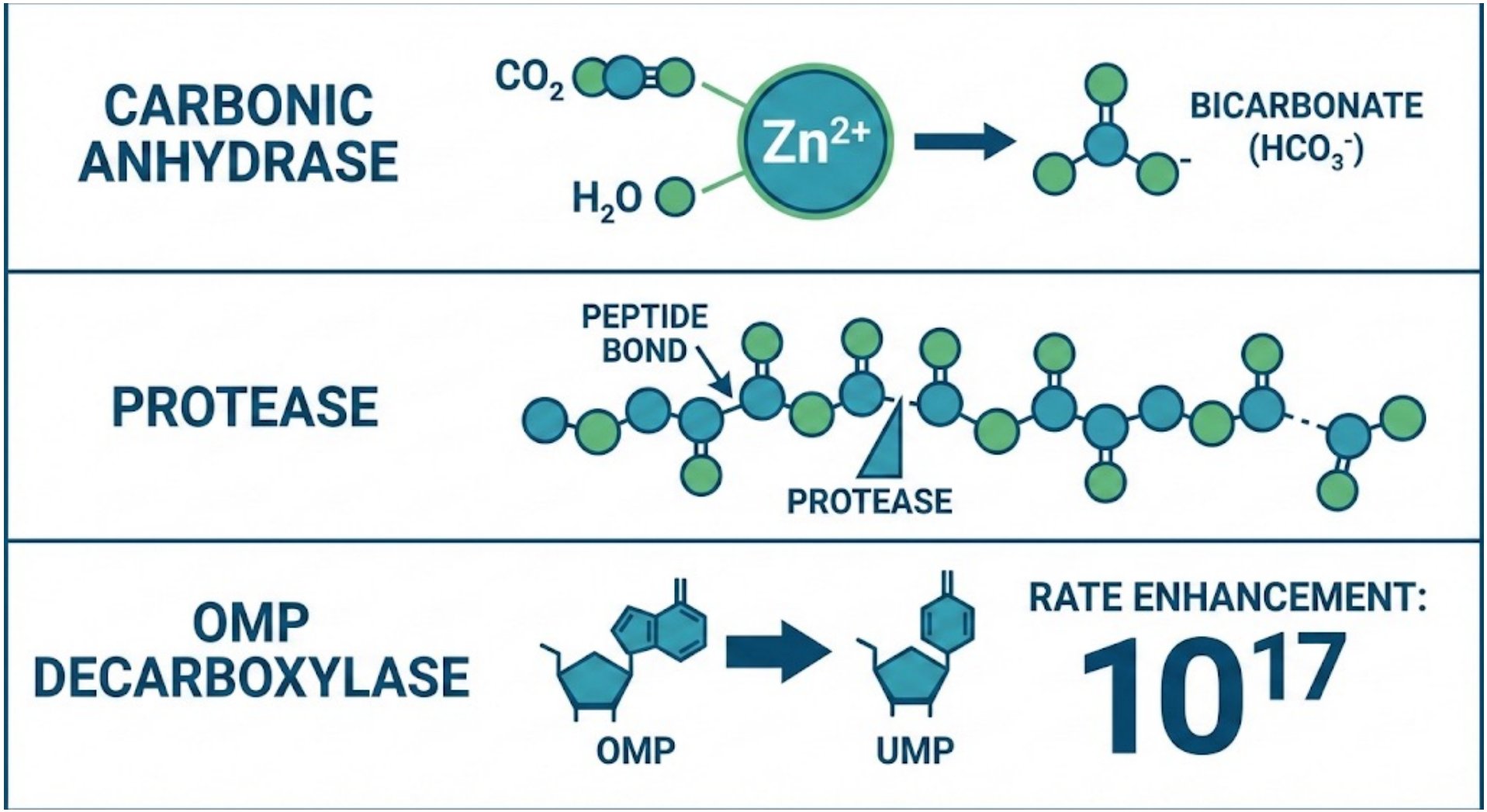
WHAT'S HAPPENING?

The answer lies at the molecular level.



By the end of this lecture, you'll understand the **drug-enzyme interaction** causing **rhabdomyolysis** and why **enzyme kinetics** is essential for every prescription.

CHAPTER 7
Basic Concepts of Enzyme Action



Chapter 7: Outline

7.1 Enzymes Are Powerful and Highly Specific Catalysts

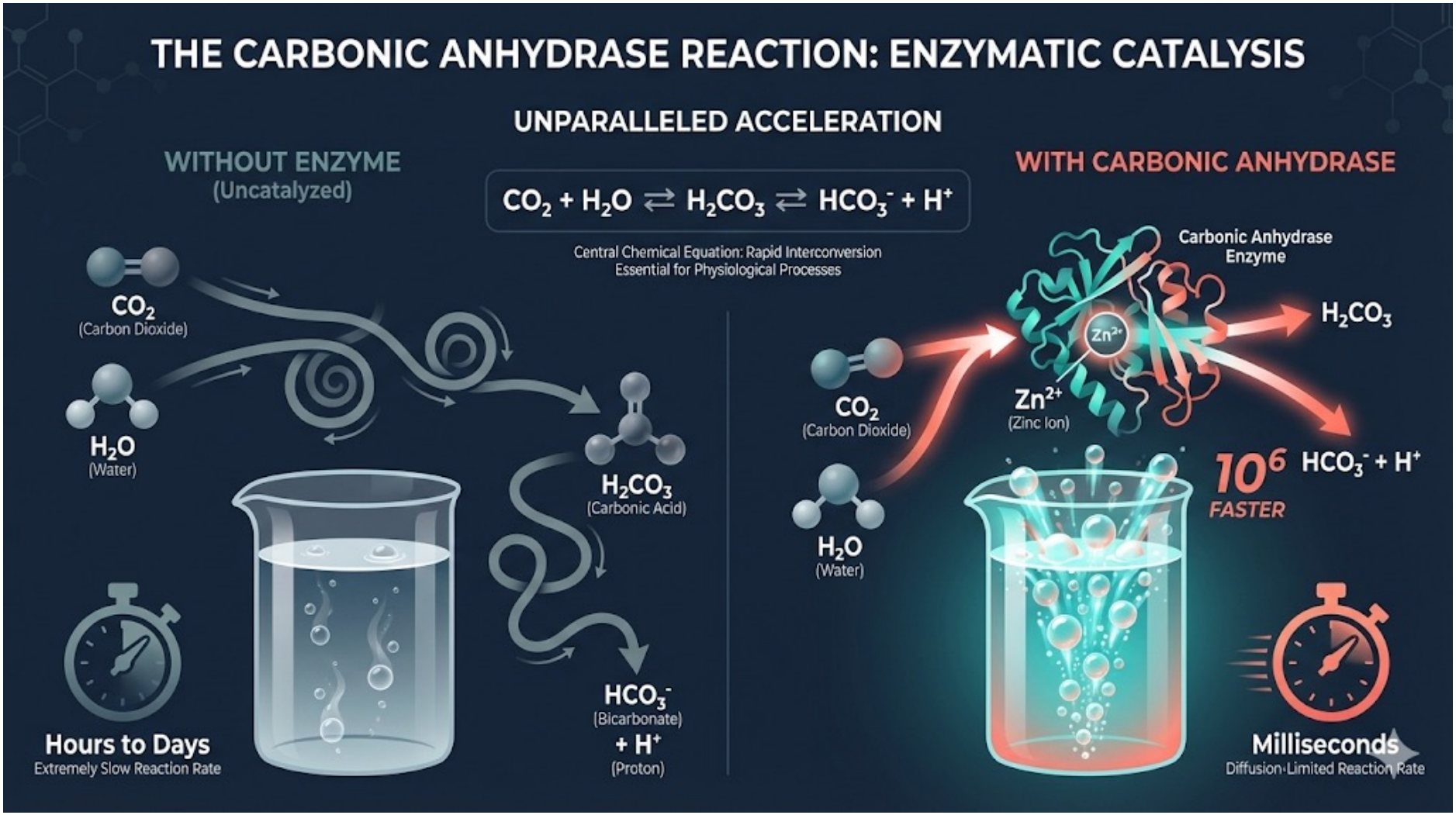
7.2 Many Enzymes Require Cofactors for Activity

7.3 Gibbs Free Energy Is a Useful Thermodynamic Function for Understanding Enzymes

7.4 Enzymes Facilitate the Formation of the Transition State

Section 7.1 Enzymes Are Powerful and Highly Specific Catalysts

- Enzymes are protein catalysts that can accelerate the rate of a reaction by factors of as much as a million or more.



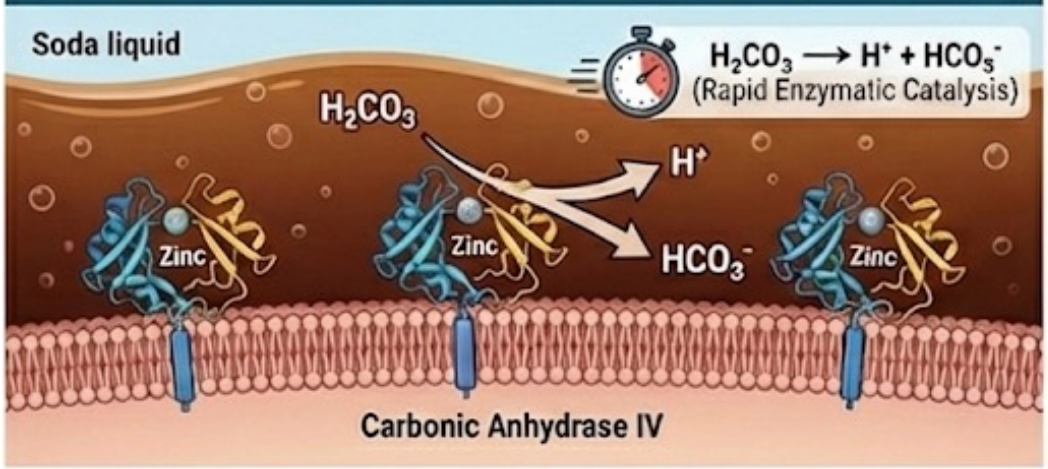
THE SODA FIZZ PHENOMENON: FROM PRESSURE TO ENZYMATIC SENSORY BURST

1. CARBONATION UNDER PRESSURE (Physics & Chemistry)

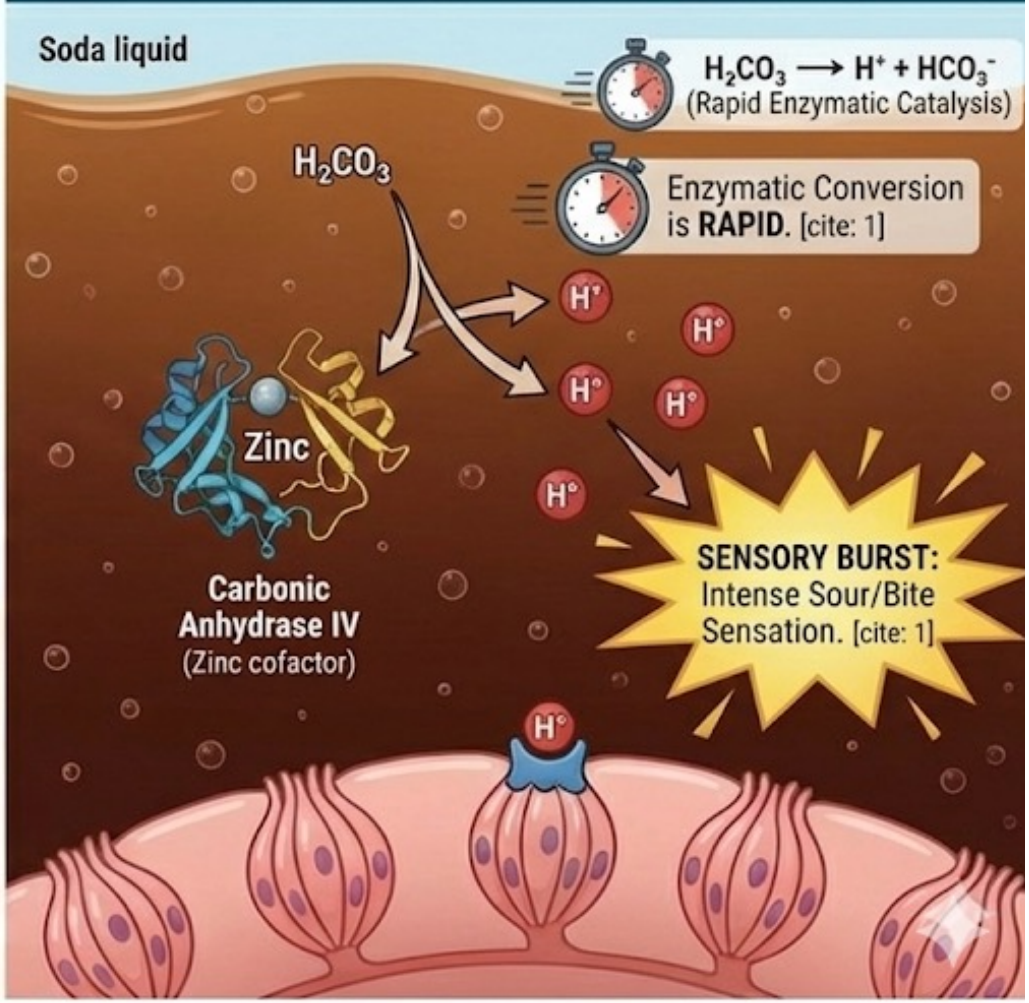
High pressure forces CO_2 into solution;
H_2CO_3 is a minor component.



3. ON TONGUE: ENZYMATIC ACCELERATION (Biochemistry)



2. ON TONGUE: ENZYMATIC ACCELERATION



How much does an enzyme speed up a reaction? A lot

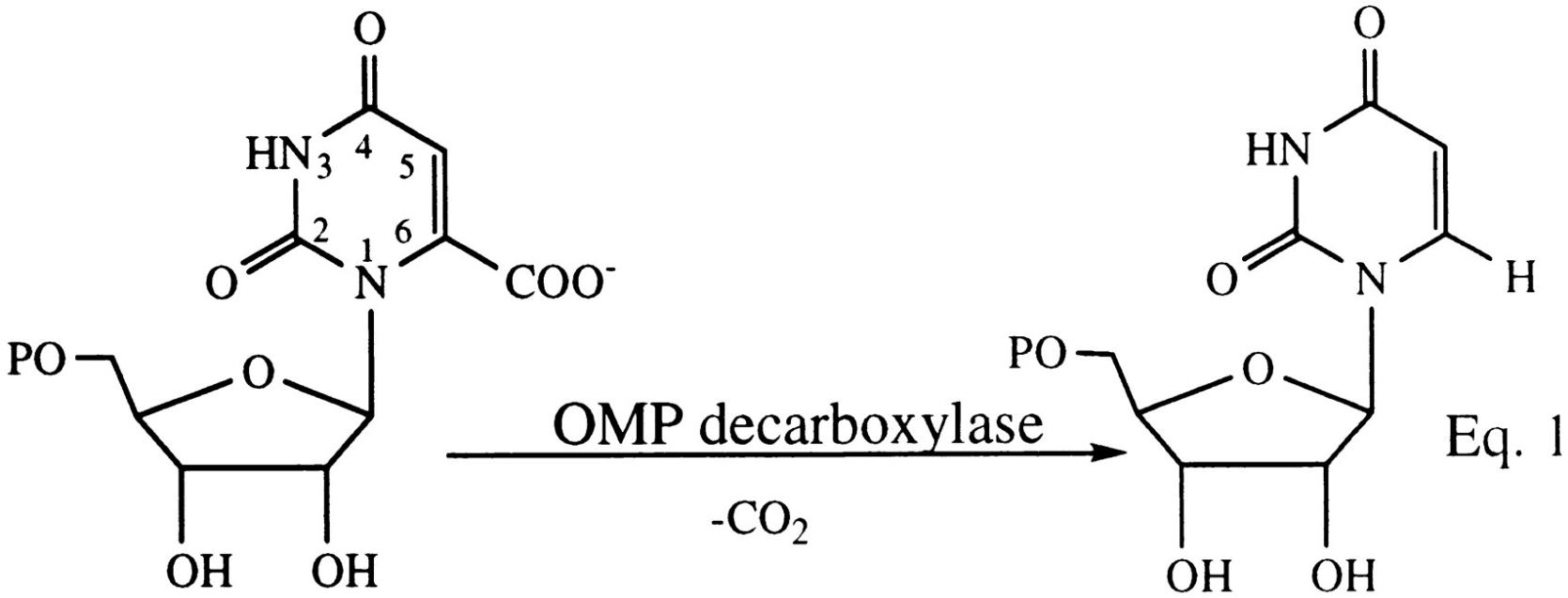


TABLE 6.1 Rate enhancement by selected enzymes

Enzyme	Nonenzymatic half-life	Uncatalyzed rate (k_{un} s^{-1})	Catalyzed rate (k_{cat} s^{-1})	Rate enhancement (k_{cat} s^{-1}/k_{un} s^{-1})
OMP decarboxylase	78,000,000 years	2.8×10^{-16}	39	1.4×10^{17}

What is the rate enhancement of an enzyme with an uncatalyzed rate of 2.9×10^{-6} s^{-1} and a catalyzed rate of 540 s^{-1}?

A. 1.0×10^{6}
B. 1.9×10^{8} ←
C. 1.9×10^{9}
D. 1.9×10^{11}



OMP decarboxylase

Table of Rate Enhancement by Selected Enzymes

TABLE 6.1 Rate enhancement by selected enzymes

Enzyme	Nonenzymatic half-life	Uncatalyzed rate (k_{un} s^{-1})	Catalyzed rate (k_{cat} s^{-1})	Rate enhancement (k_{cat} s^{-1}/k_{un} s^{-1})
OMP decarboxylase	78,000,000 years	2.8×10^{-16}	39	1.4×10^{17}
Staphylococcal nuclease	130,000 years	1.7×10^{-13}	95	5.6×10^{14}
AMP nucleosidase	69,000 years	1.0×10^{-11}	60	6.0×10^{12}
Carboxypeptidase A	7.3 years	3.0×10^{-9}	578	1.9×10^{11}
Ketosteroid isomerase	7 weeks	1.7×10^{-7}	66,000	3.9×10^{11}
Triose phosphate isomerase	1.9 days	4.3×10^{-6}	4300	1.0×10^{9}
Chorismate mutase	7.4 hours	2.6×10^{-5}	50	1.9×10^{6}
Carbonic anhydrase	5 seconds	1.3×10^{-1}	1×10^{6}	7.7×10^{6}

Abbreviations: OMP, orotidine monophosphate; AMP, adenosine monophosphate. Source: Data from A. Radzicka and R. Wolfenden, *Science* 267:90–93, 1995.

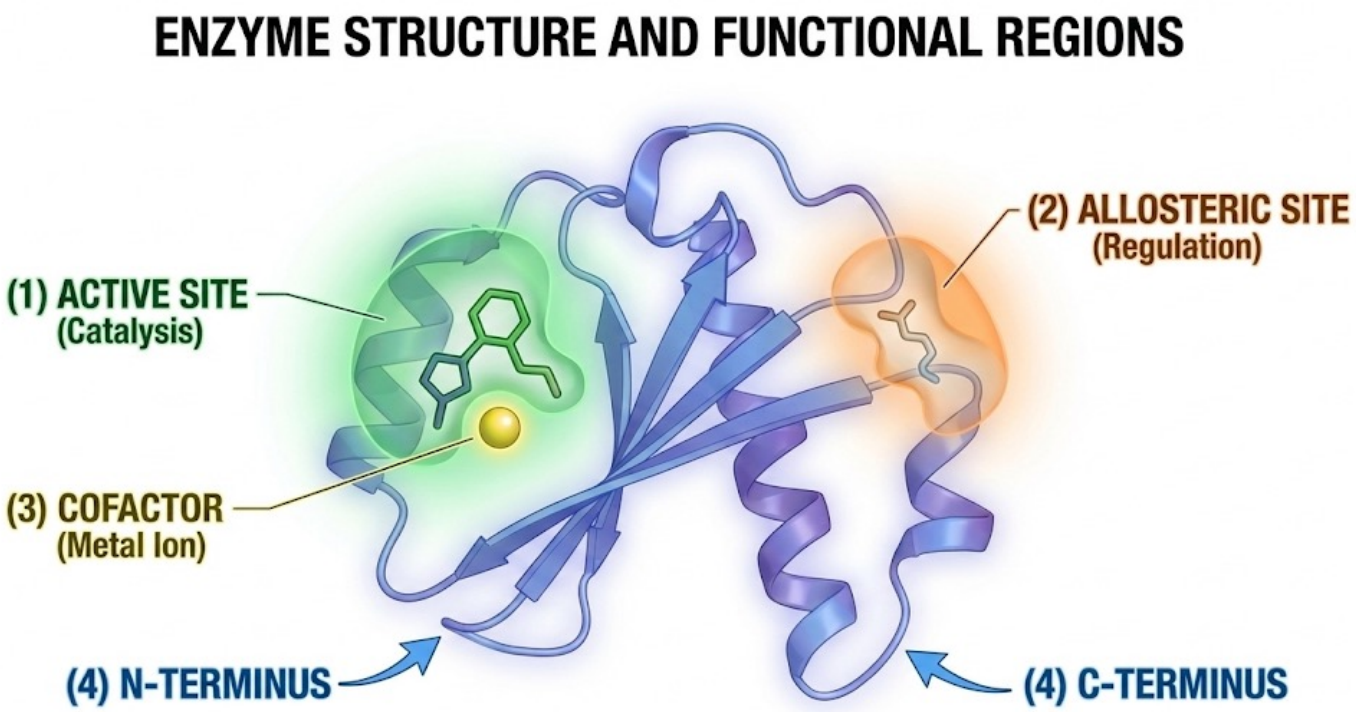
Tymoczko et al., *Biochemistry: A Short Course*, 4e, © 2019 W. H. Freeman and Company

Catalysts speed up chemical reactions without being consumed by them.

Enzymes are powerful biological catalysts.

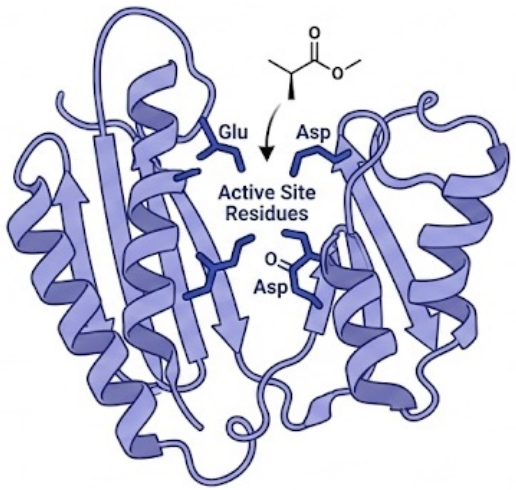
- They are made by all living organisms.
- They dramatically enhance and control the rates of chemical reactions.
- They are almost all proteins.

The discovery of catalytically active RNA molecules, called **ribozymes**, provides compelling evidence that RNA was a biocatalyst early in evolution.

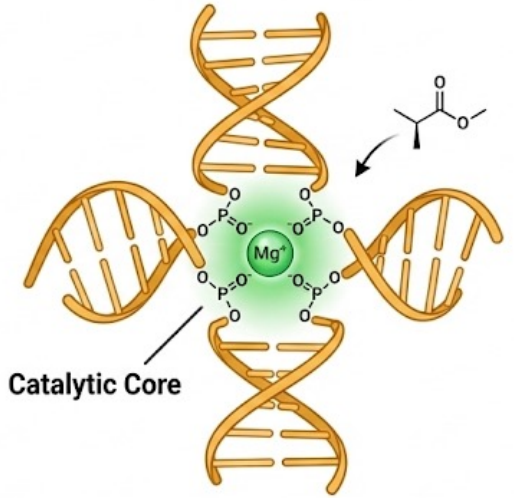


CATALYTIC COMPARISON: PROTEIN vs. RNA

PROTEIN ENZYME
(Lysozyme)



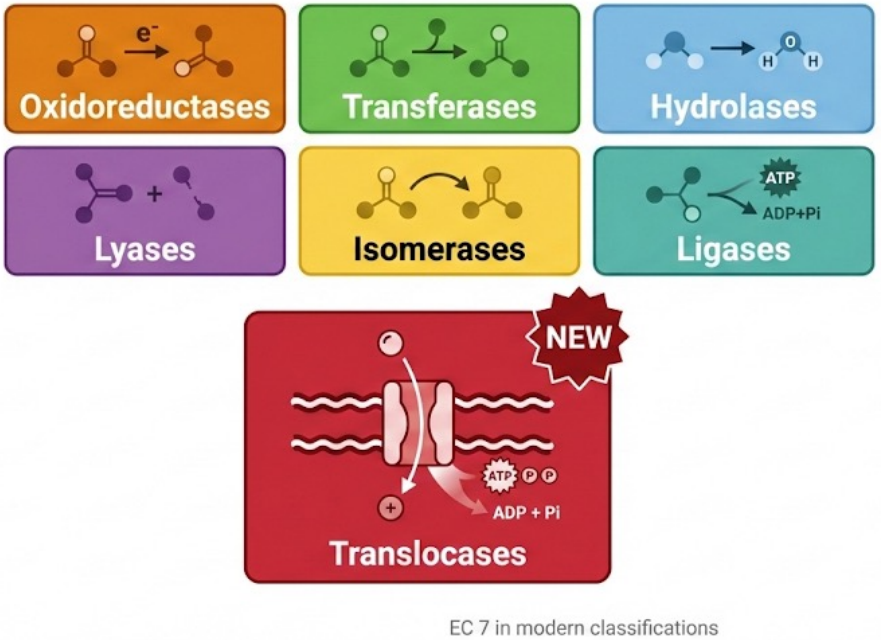
RIBOZYME
(Hammerhead)



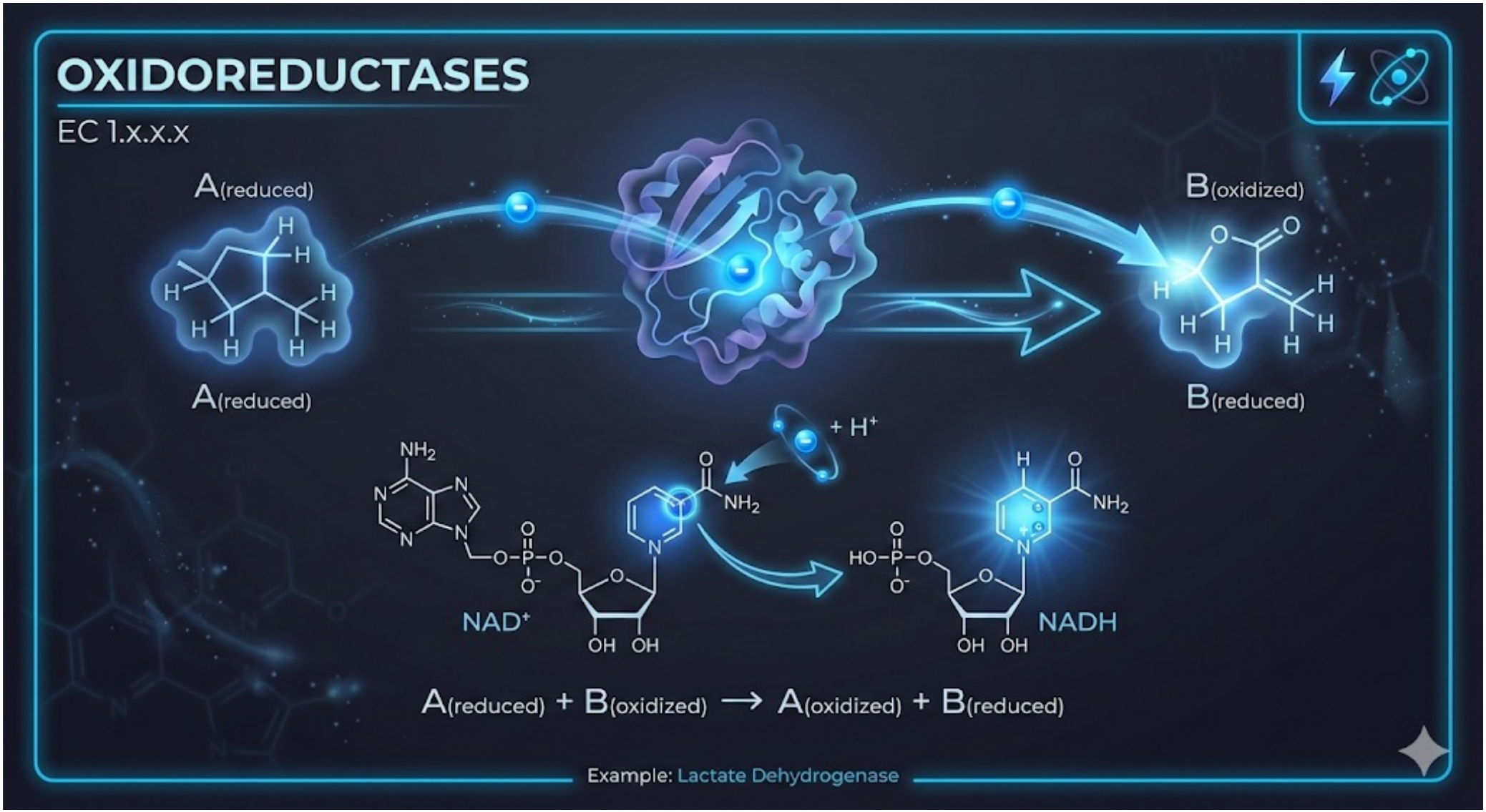
'Catalysis requires precise 3D positioning—not necessarily protein.'

There Are ~~Six~~ 7 Major Classes of Enzymes

Class	Type of reaction
1. Oxidoreductases	Oxidation–reduction
2. Transferases	Group transfer
3. Hydrolases	Hydrolysis reactions (transfer of functional groups to water)
4. Lyases	Addition or removal of groups to form double bonds
5. Isomerases	Isomerization (intramolecular group transfer)
6. Ligases	Ligation of two substrates at the expense of ATP hydrolysis
7. Translocases	Movement of ions or molecules across membranes or within membranes



EC 7 in modern classifications

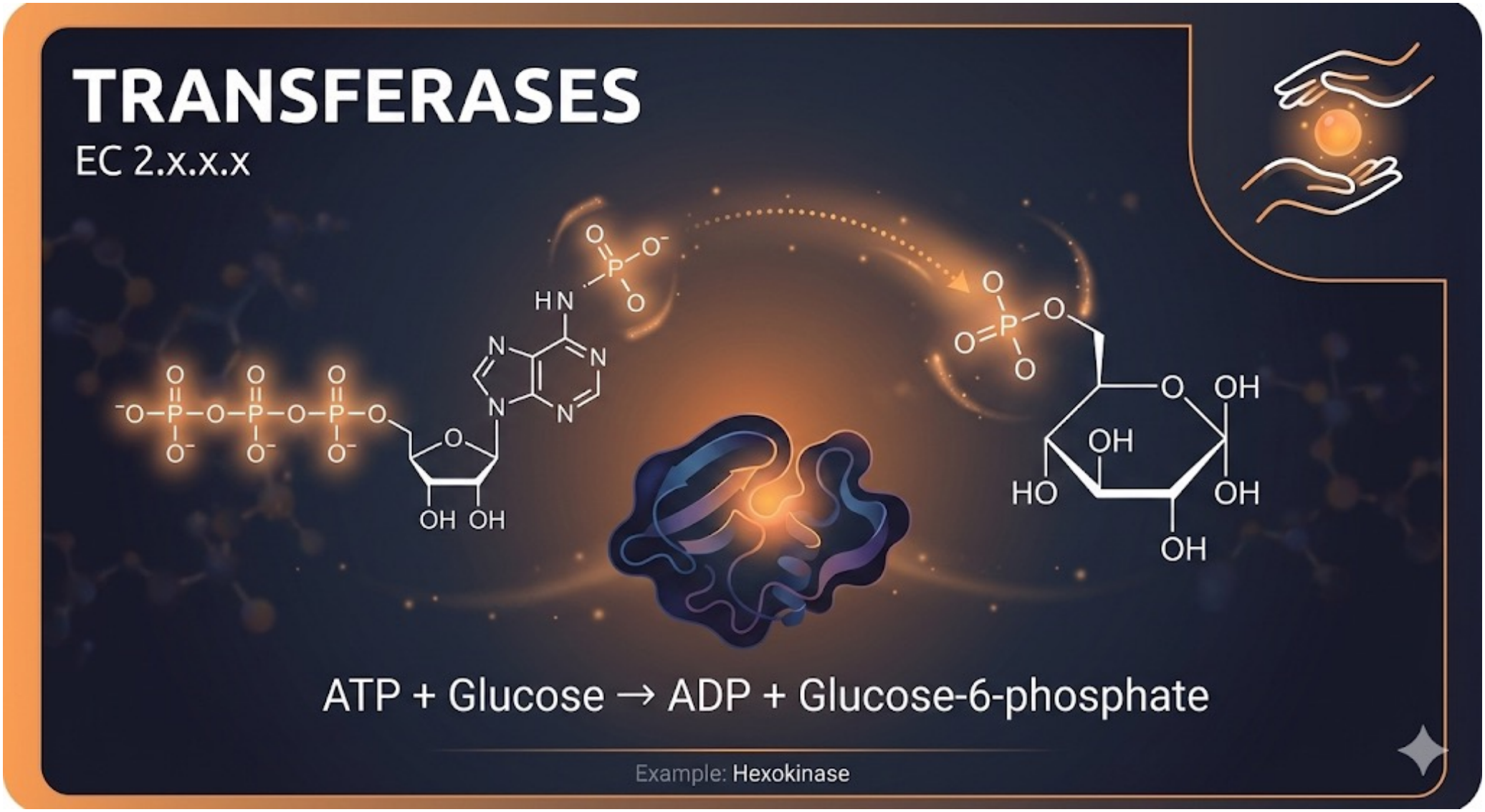


These enzymes transfer electrons between molecules. In other words, these enzymes catalyze oxidation–reduction reactions.

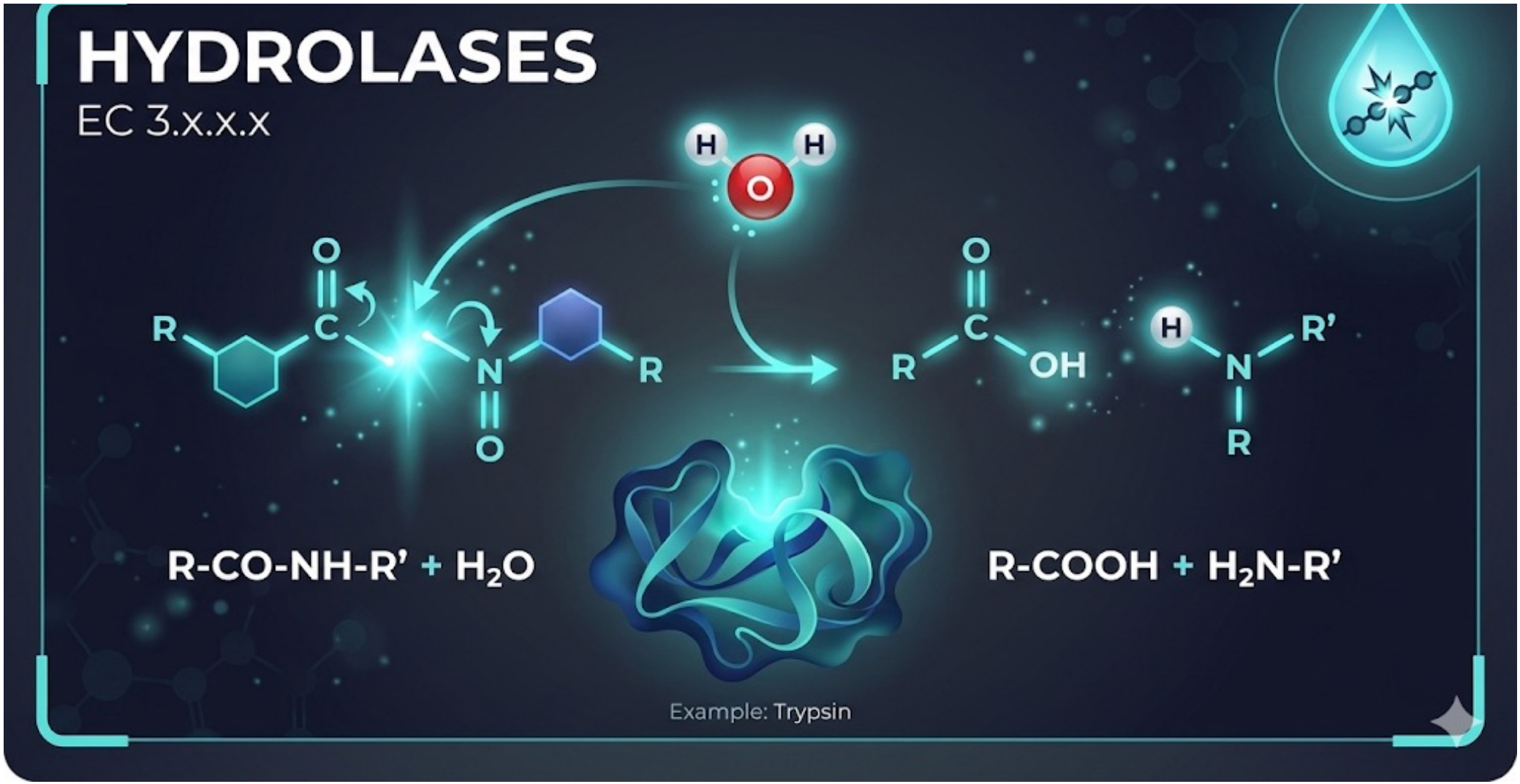
The Importance of Oxidoreductase: Elephant toothpaste

- **A Fun Example of an oxidoreductase: Catalase**
- Catalase is present in the peroxisomes of eukaryotic cells and serves to detoxify hydrogen peroxide (H_2O_2), a potentially harmful by-product of cellular metabolism. The enzyme catalyzes the decomposition of hydrogen peroxide into water and oxygen gas: 2H2O2→2H2O+O22H2O2→2H2O+O2
- **Fun Experiment:** One fun demonstration involving catalase is the "Elephant Toothpaste" experiment. When a sample rich in catalase (like liver puree or yeast solution) is mixed with hydrogen peroxide and a drop of dish soap, the rapid decomposition of hydrogen peroxide produces oxygen gas. The soap catches this oxygen, creating a large foam resembling toothpaste big enough for an "elephant."



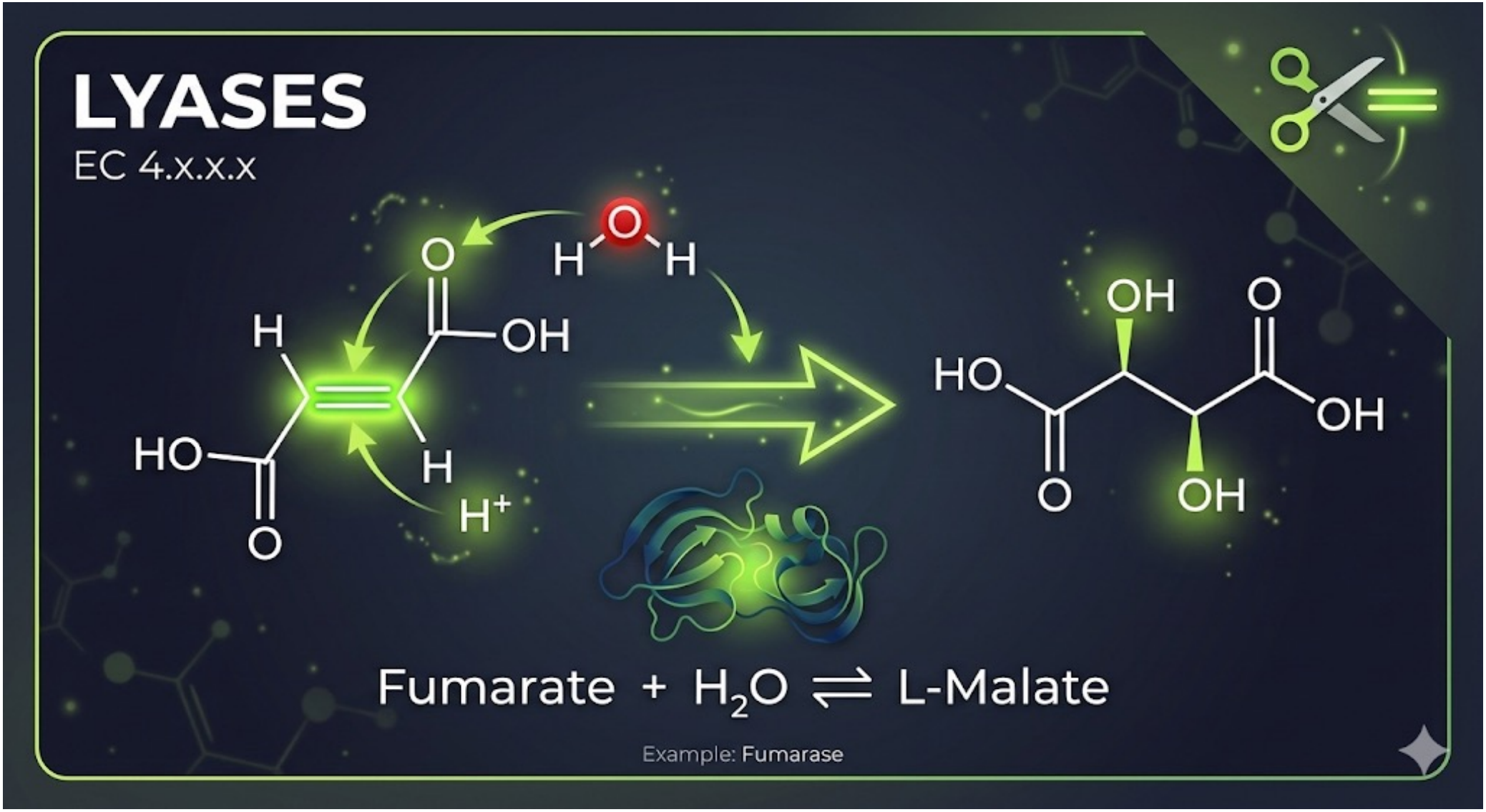


These enzymes transfer functional groups between molecules. Aminotransferases are prominent in amino acid synthesis and degradation, where they shuffle amine groups between donor and acceptor molecules.

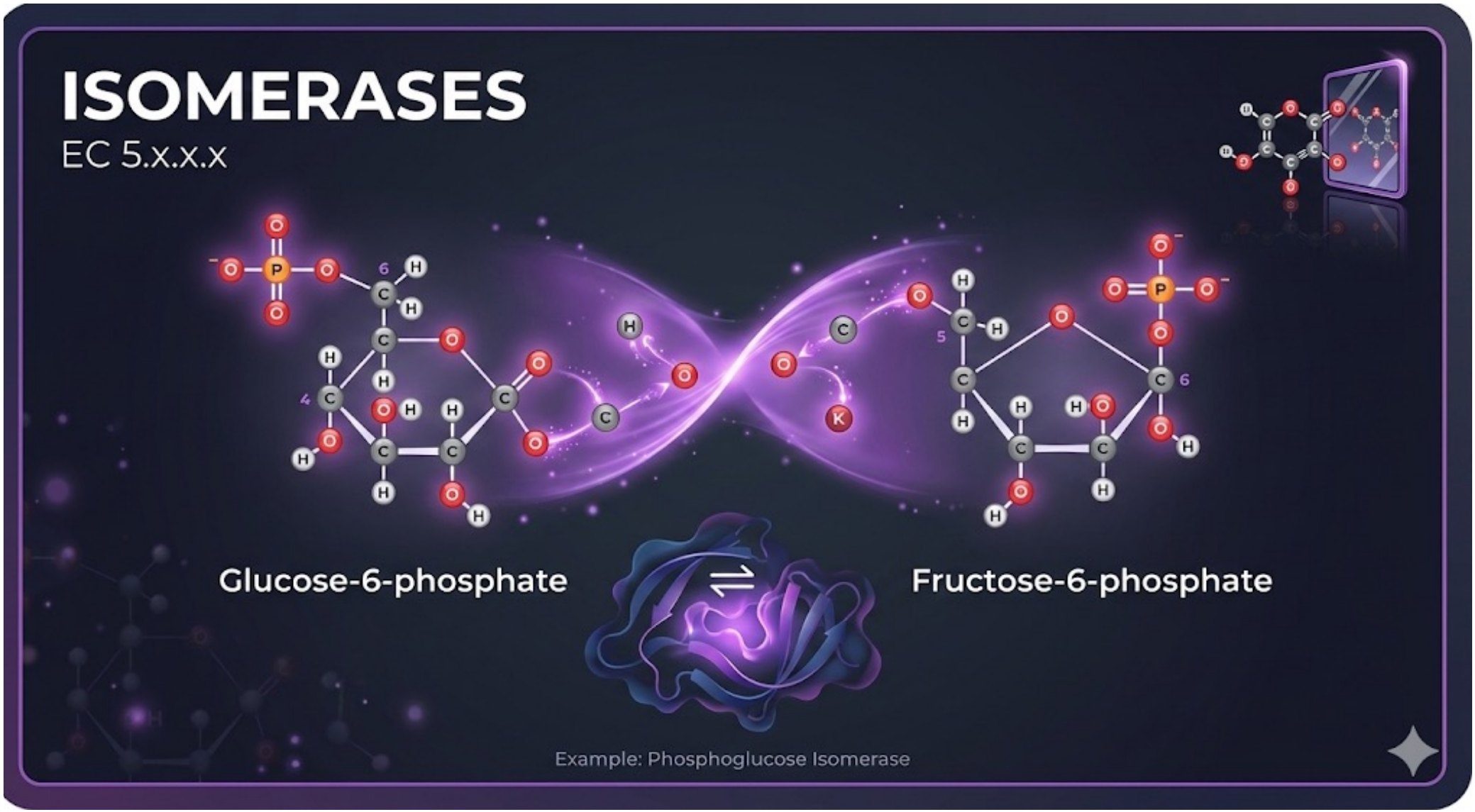


A hydrolase cleaves molecules by the addition of water. Trypsin, the proteolytic enzyme already discussed, is a hydrolyase.

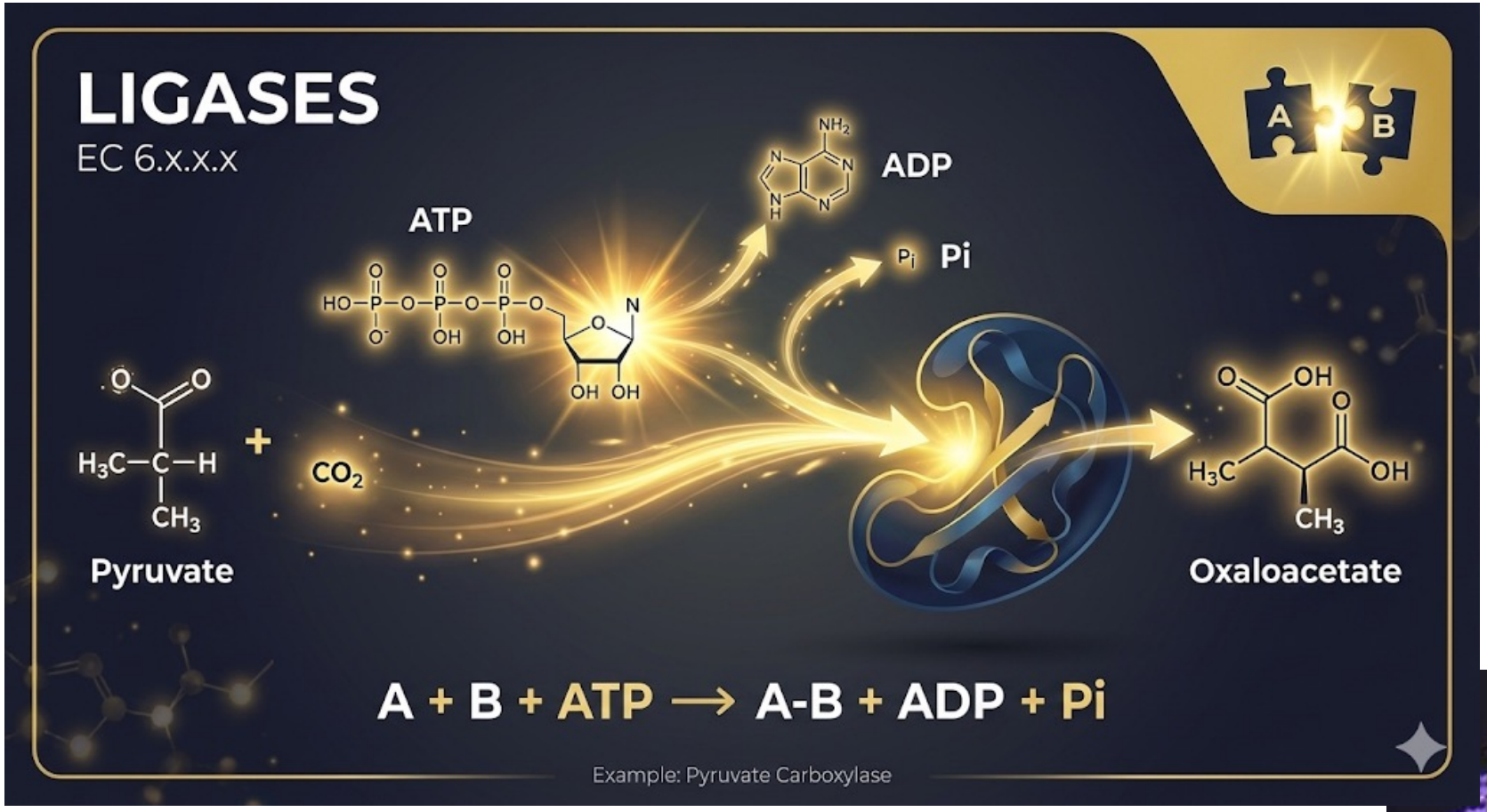
Lactaid
Still Down For Dairy



A lyase adds atoms or functional groups to a double bond or removes them to form double bonds. Does not usually requires extra energy.



These enzymes move functional groups within a molecule. We will meet triose phosphate isomerase in glycolysis.



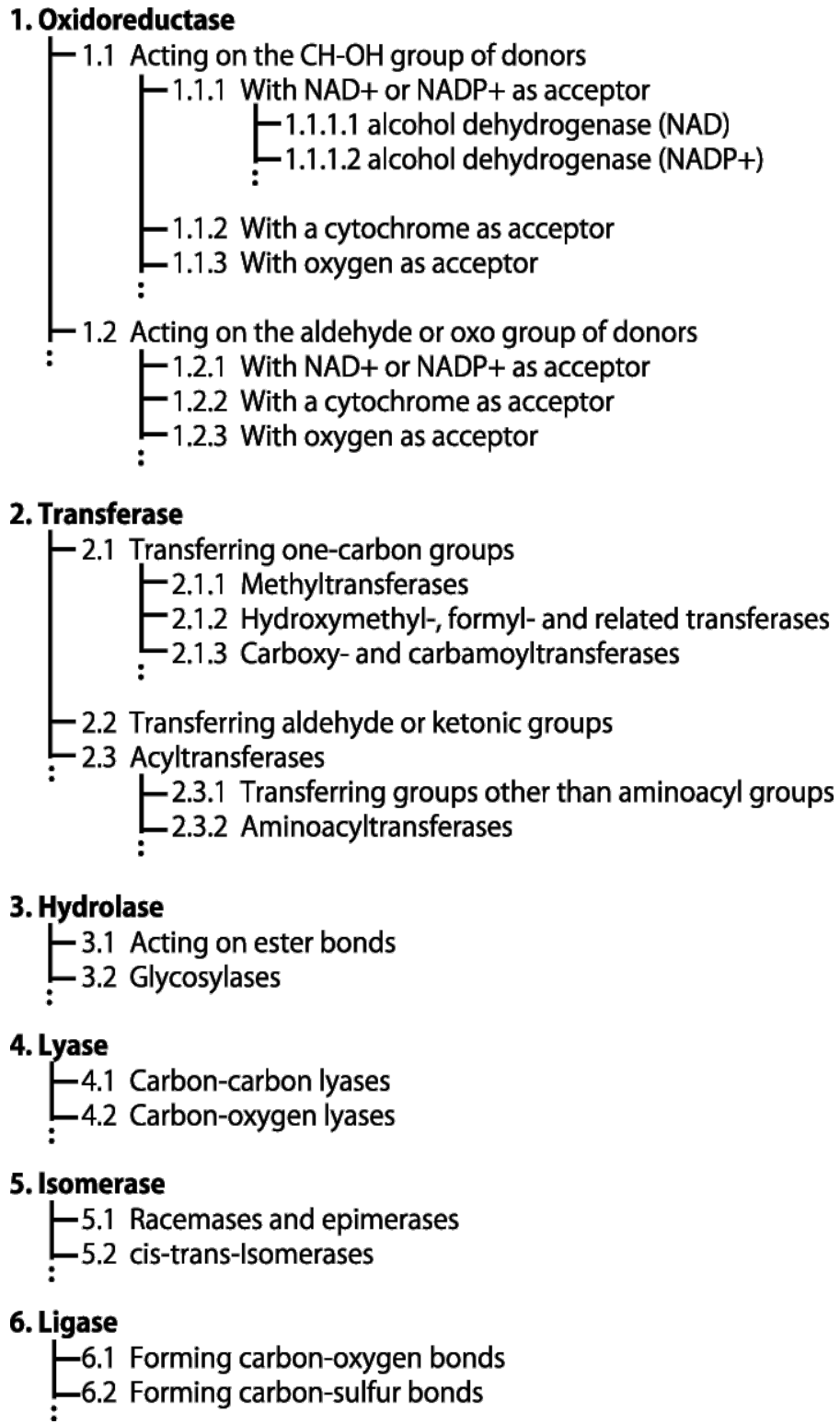
Ligases join two molecules in a reaction powered by ATP hydrolysis. DNA ligase, an important enzyme in DNA replication, is representative of this class. ATP synthase shows that enzyme classification is based on the reaction mechanism, not physiological function.

$$\mathrm{ADP} + \mathrm{P}_i + \mathrm{H}^+ \xrightarrow{\text{ATP Synthase}} \mathrm{ATP} + \mathrm{H_2O}$$



ATP Synthase

EC naming convention

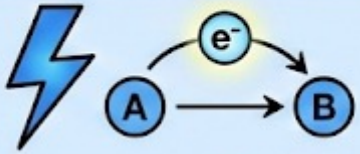


THE SIX ENZYME CLASSES

Old Tiresome Harold Loves Indoor Lamps

Class 1: OXIDOREDUCTASES

O



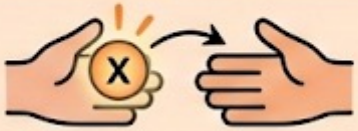
A(red) + B(ox) ⟶ A(ox) + B(red)

Example: Lactate dehydrogenase

Key phrase: Electron transfer.

Class 2: TRANSFERASES

T



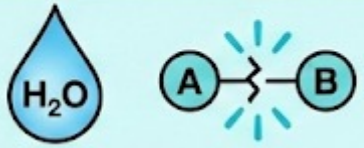
A-X + B ⟶ A + B-X

Example: Hexokinase

Key phrase: Group transfer.

Class 3: HYDROLASES

H



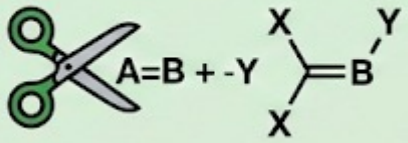
A-B + H_2O ⟶ A-OH + H-B

Example: Trypsin

Key phrase: Water breaks bonds.

Class 4: LYASES

L



A=B + X-Y ⇄ A(X)-B(Y)

Example: Fumarase

Key phrase: Add/remove from double bonds.

Class 5: ISOMERASES

I



A ⇄ A' ⟶ A' (same formula, different arrangement)

Example: Phosphoglucose isomerase

Key phrase: Rearrangement

Class 6: LIGASES

L

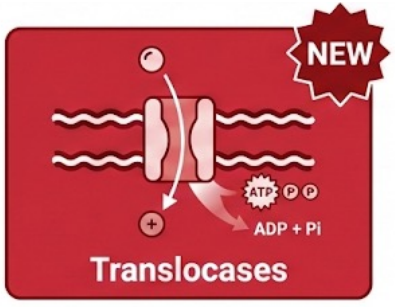


A + B + ATP ⟶ A-B + ADP + Pi

Example: DNA ligase

Key phrase: Join with ATP energy.

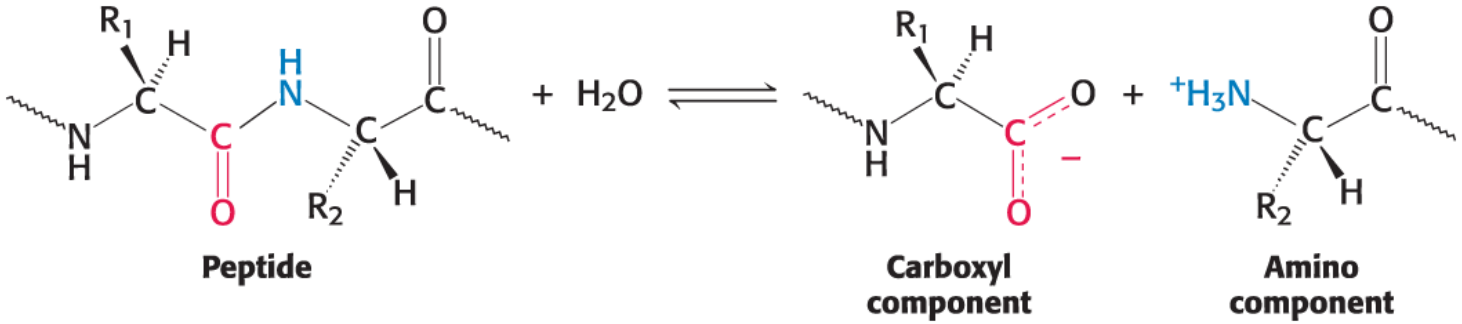
FOOTER | EC Number: 1.x.x.x = Oxidoreductase, 2.x.x.x = Transferase, 3.x.x.x = Hydrolase, 4.x.x.x = Lyase, 5.x.x.x = Isomerase, 6.x.x.x = Ligase



EC 7 in modern classifications

Proteases

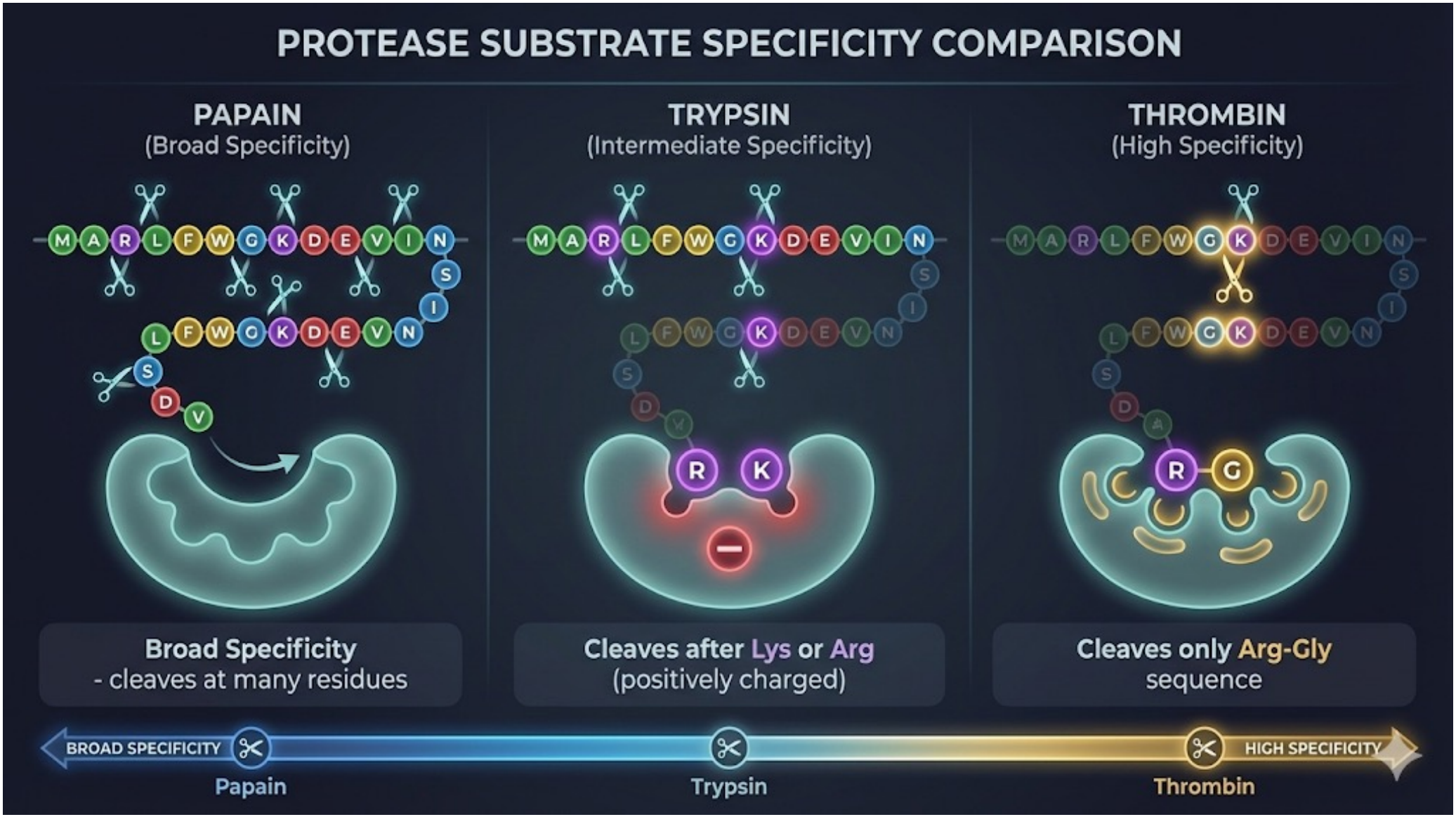
Proteases (**proteolytic enzymes**) catalyze proteolysis, the hydrolysis of a peptide bond.



All proteases break a chemical bond by the addition of a water molecule.

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Diagram of Enzyme Specificity



Section 7.2 Many Enzymes Require Cofactors for Activity

- Cofactors are small molecules that some enzymes require for activity. **The two main classes of cofactors** are coenzymes—organic molecules derived from vitamins—and metals.
- Tightly bound coenzymes are called prosthetic groups.
- An enzyme with its cofactor is a holoenzyme. Without the cofactor, the enzyme is called an apoenzyme.

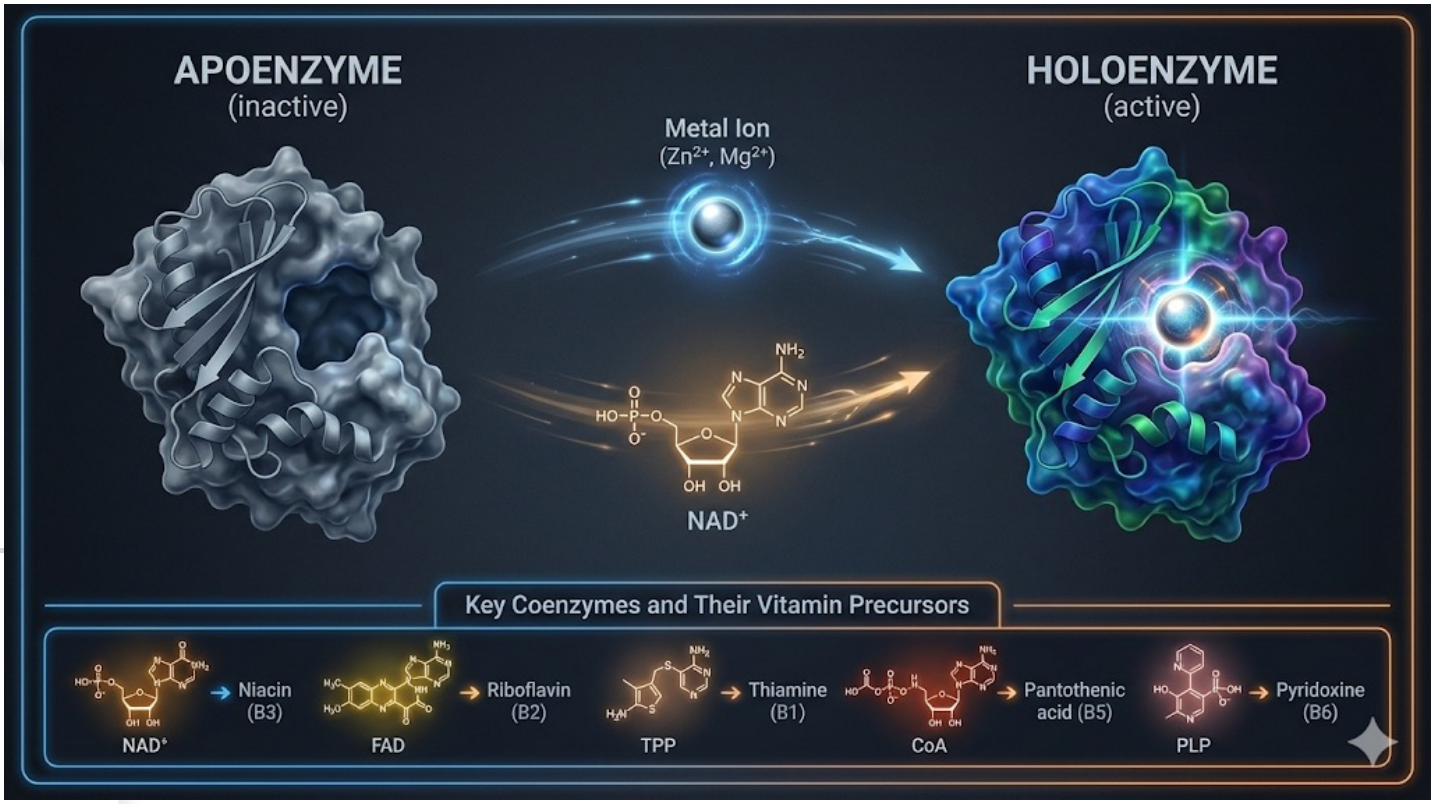


TABLE 6.2 Enzyme cofactors

Cofactor	Enzyme*
Coenzyme†	
Thiamine pyrophosphate (TPP)	Pyruvate dehydrogenase
Flavin adenine dinucleotide (FAD)	Monoamine oxidase
Nicotinamide adenine dinucleotide (NAD^+)	Lactate dehydrogenase
Pyridoxal phosphate (PLP)	Glycogen phosphorylase
Coenzyme A (CoA)	Acetyl CoA carboxylase
Biotin	Pyruvate carboxylase
5′-Deoxyadenosylcobalamin	Methylmalonyl mutase
Tetrahydrofolate	Thymidylate synthase
Metal	
Zn^{2+}	Carbonic anhydrase
Mg^{2+}	*Eco*RV
Ni^{2+}	Urease
Mo	Nitrogenase
Se	Glutathione peroxidase
$Mn^{2+\leftrightarrow 3+}$	Superoxide dismutase
K^+	Acetoacetyl CoA thiolase

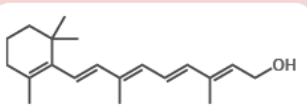
*The enzymes listed are examples of enzymes that employ the indicated cofactor.
† Often derived from vitamins, coenzymes can be either tightly or loosely bound to the enzyme.

Tymoczko et al., *Biochemistry: A Short Course*, 4e, © 2019 W. H. Freeman and Company

THE CHEMICAL STRUCTURES OF VITAMINS

Vitamins are the essential nutrients that our body needs in small amounts. More specifically, an organic compound is defined as a vitamin when it is required by an organism, but not synthesised by that organism in the required amounts (or at all). There are thirteen recognised vitamins.

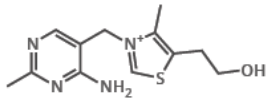
VITAMIN A



RETINOL
active form in mammalian tissues

Important for eyesight. Also strengthens immune system and keeps skin and linings of parts of the body healthy.

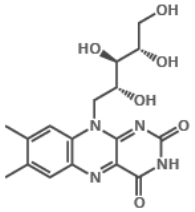
VITAMIN B1



THIAMIN
can also occur in pyrophosphate ester form

Used to keep nerves & muscle tissue healthy. Also important for processing of carbohydrates and some proteins.

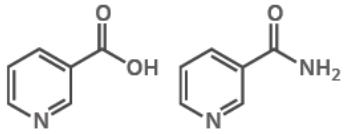
VITAMIN B2



RIBOFLAVIN
excess turns urine bright yellow

Important for body growth, red blood cell production, and keeping the eyes healthy. Also helps processing of carbohydrates.

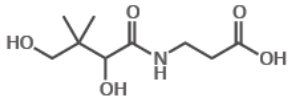
VITAMIN B3



NICOTINIC ACID **NICOTINEAMIDE**
niacin is collective name for these compounds

Helps with digestion and digestive system health. Also helps with the processing of carbohydrates.

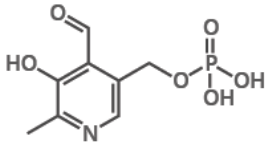
VITAMIN B5



PANTOTHENIC ACID
also occurs in pyrophosphate ester form

Important for manufacturing red blood cells and maintaining a healthy digestive system. Also helps process carbohydrates.

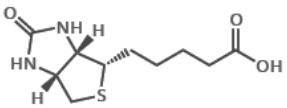
VITAMIN B6



PYRIDOXAL PHOSPHATE
active form in mammalian tissues

Helps make some brain chemicals; needed for normal brain function. Also helps make red blood cells and immune system cells.

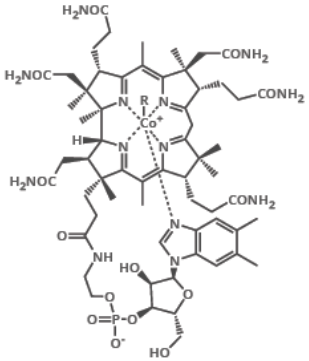
VITAMIN B7



BIOTIN
produced by intestinal bacteria

Needed for metabolism of various compounds. Often recommended for strengthening hair, but evidence is variable.

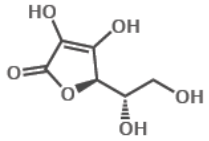
VITAMIN B12



COBALAMIN
usually contains CN as the R group

Important for the nervous system, for making red blood cells, and helps in the production of DNA and RNA.

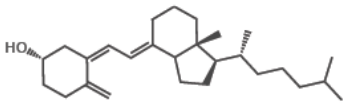
VITAMIN C



ASCORBIC ACID
deficiency can cause scurvy

Important for a healthy immune system; helps produce collagen, used to make skin and other tissues. Also helps wound healing.

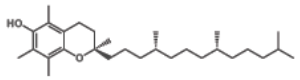
VITAMIN D



CHOLECALCIFEROL
natural form; different form used in supplements

Important for bone health and maintaining the immune system function. May also have a preventative role in cancers.

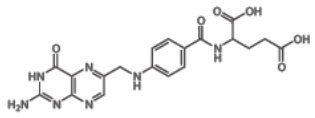
VITAMIN E



ALPHA-TOCOPHEROL
group includes tocopherols & tocotrienols

An antioxidant that helps prevent damage to cells and may have a preventative role in cancer. Also helps make red blood cells.

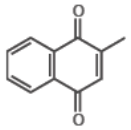
VITAMIN B9



FOLIC ACID
found as tetrahydrofolate in food

Important for brain function & mental health. Aids production of DNA & RNA. Important when tissues are growing quickly.

VITAMIN K



MENADIONE
all K vitamins are menadione or derivatives

Helps blood clot properly, & plays a key role in bone health. Newborns receive vitamin K injections to prevent bleeding.

Key

Vitamins can be divided broadly into two classes.

WATER-SOLUBLE VITAMINS
These vitamins are not stored in the body. As such, generally, they are required more frequently than the fat-soluble vitamins.

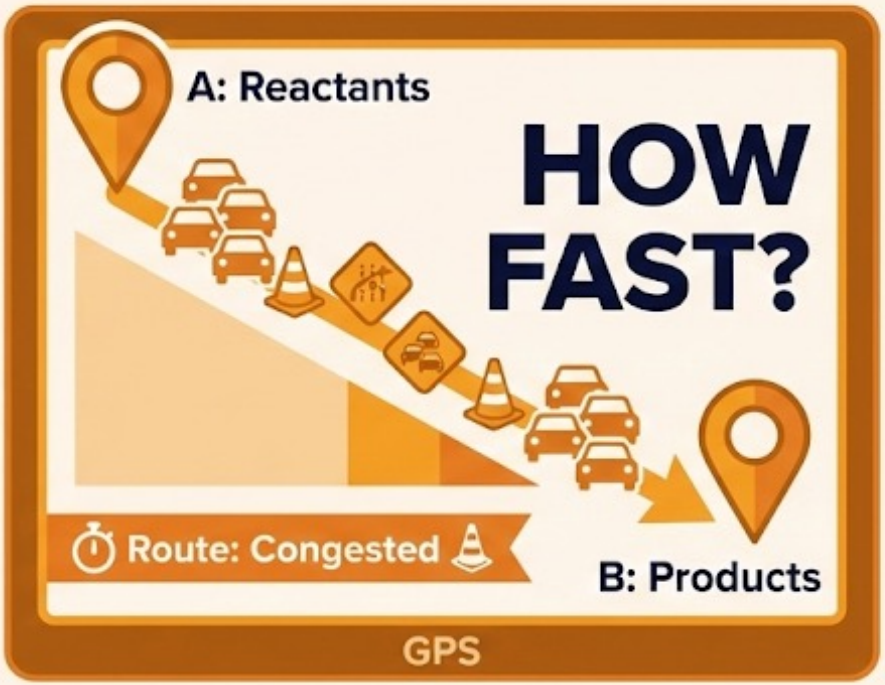
FAT-SOLUBLE VITAMINS
These vitamins are stored in the liver and fatty tissues until required. As such, they can be harmful if too much is taken in.

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Before We Talk About HOW Enzymes Work, Let's Ask: What CAN Enzymes Do?



VS.



THERMODYNAMICS

ΔG (Thermodynamics)

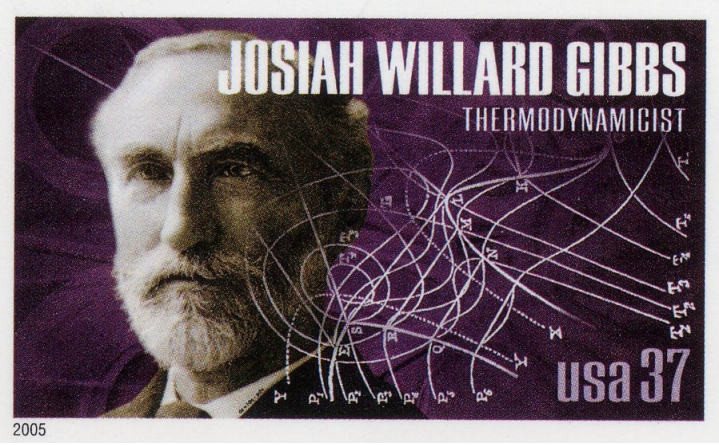
KINETICS

ΔG‡ (Kinetics)

Enzymes answer the second question—not the first.

Let’s forget about enzymes for a second and just think about reactions.

Section 7.3 Gibbs Free Energy is a Useful Thermodynamic Function for Understanding Enzymes



The properties of chemical reactions and whether they can take place at all depend on free-energy differences.

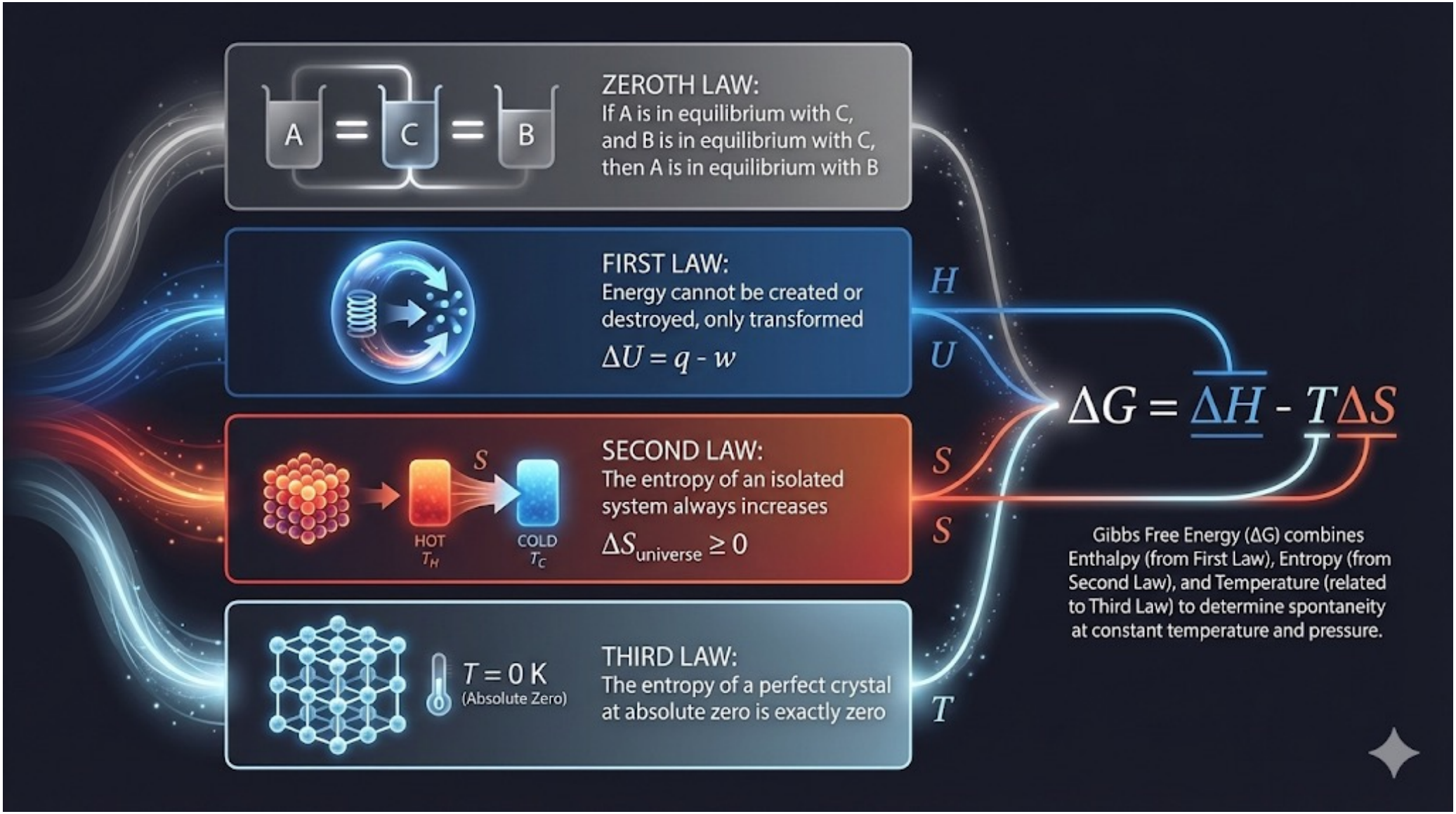
Gibbs free energy, or **free energy** (***G***), is a thermodynamic property that is a measure of the energy that is capable of doing work.

To understand how enzymes operate, we need to consider two thermodynamic properties of reactions:

1. The free-energy difference (ΔG) between the products and the reactants
   - Determines whether the reaction will take place spontaneously.
2. The free energy required to initiate the conversion of reactants into products
   - Determines the rate of the reaction
   - Enzymes affect only this property

Free energy (G) is a measure of energy capable of doing work. The change in free energy when a reaction occurs is symbolized by ΔG.

Enzymes do not alter the ΔG of a reaction.



Gibb’s free energy equation describes thermodynamic processes

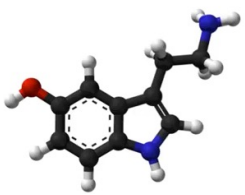
Gibb's free energy equation

$$G = H - TS$$

free energy — enthalpy — temperature x entropy

Enthalpy (H)
Total energy of the system
(energy in bonds)

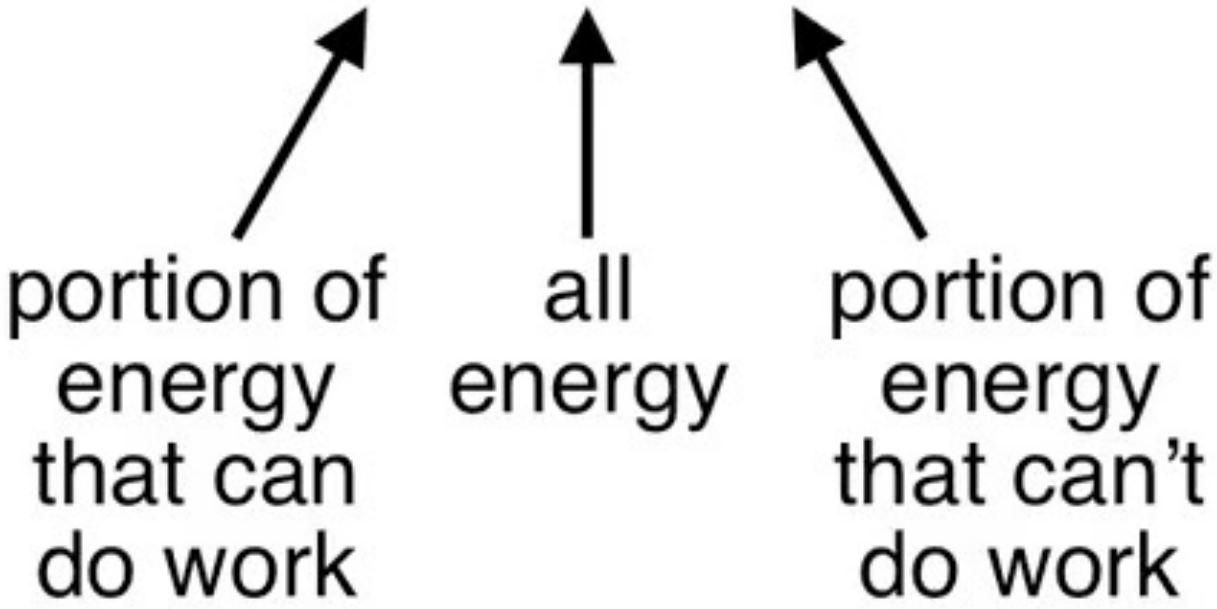
G = **H** - TS



Randomness in the system

Free energy (G)
(energy that can do work)

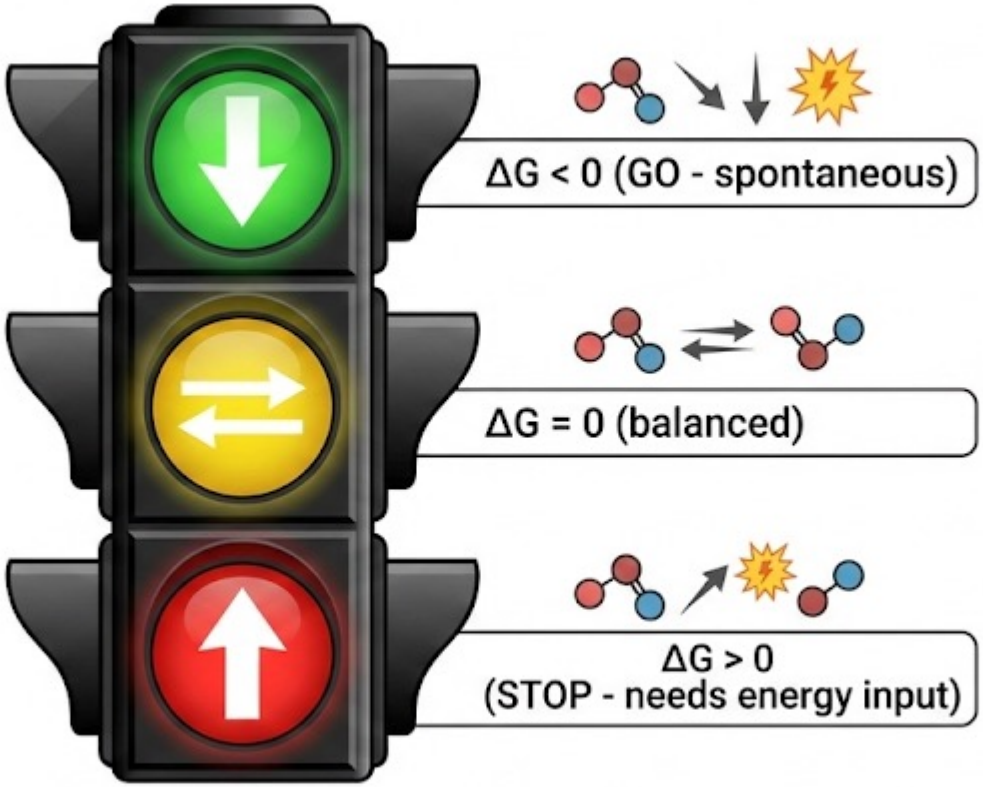
$$G = H - TS$$



How do we tell if reaction is:

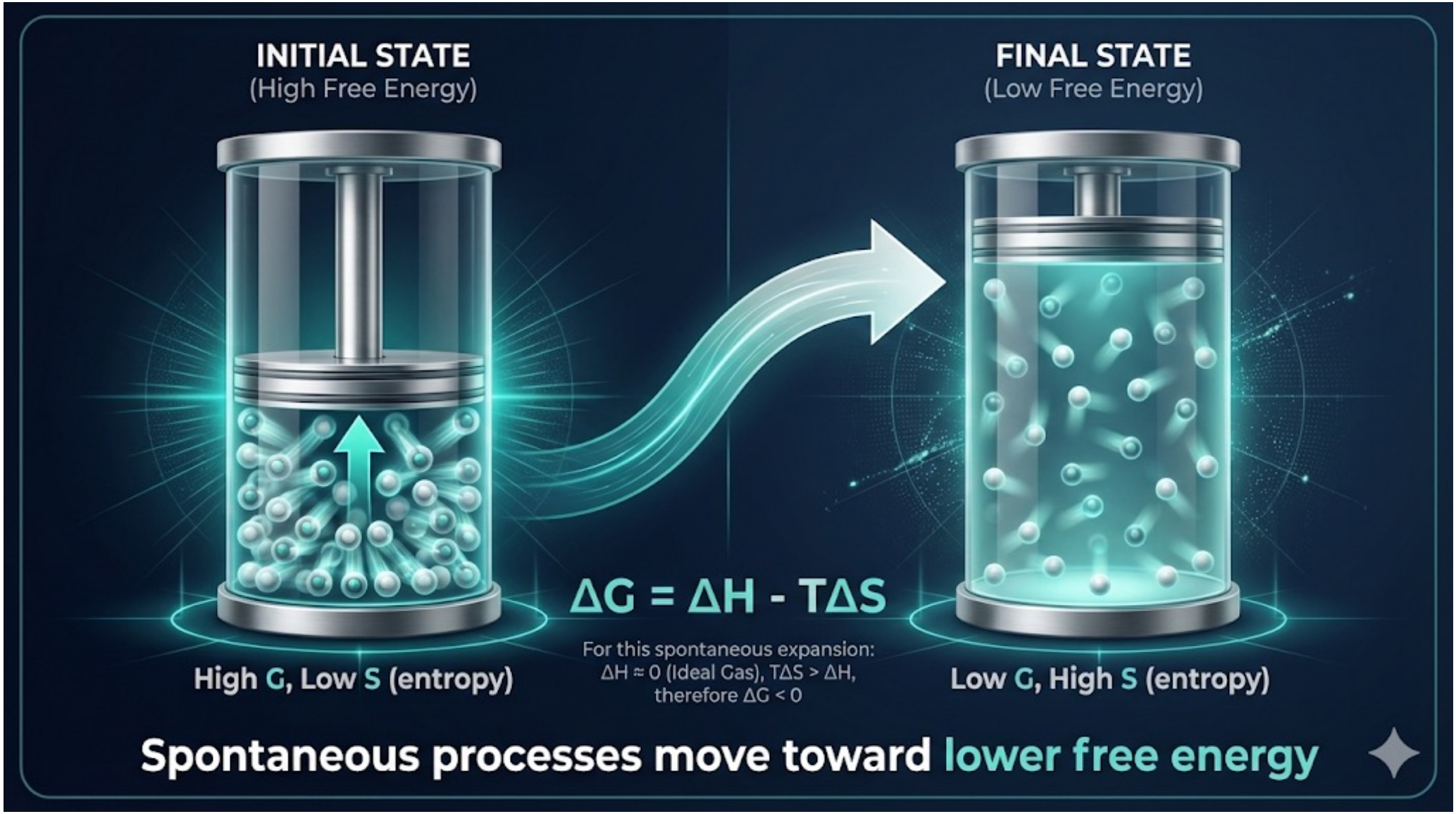
1. Spontaneous or not
2. Equilibrium constant
3. Directionality
4. Velocity

THE THERMODYNAMIC TRAFFIC LIGHT

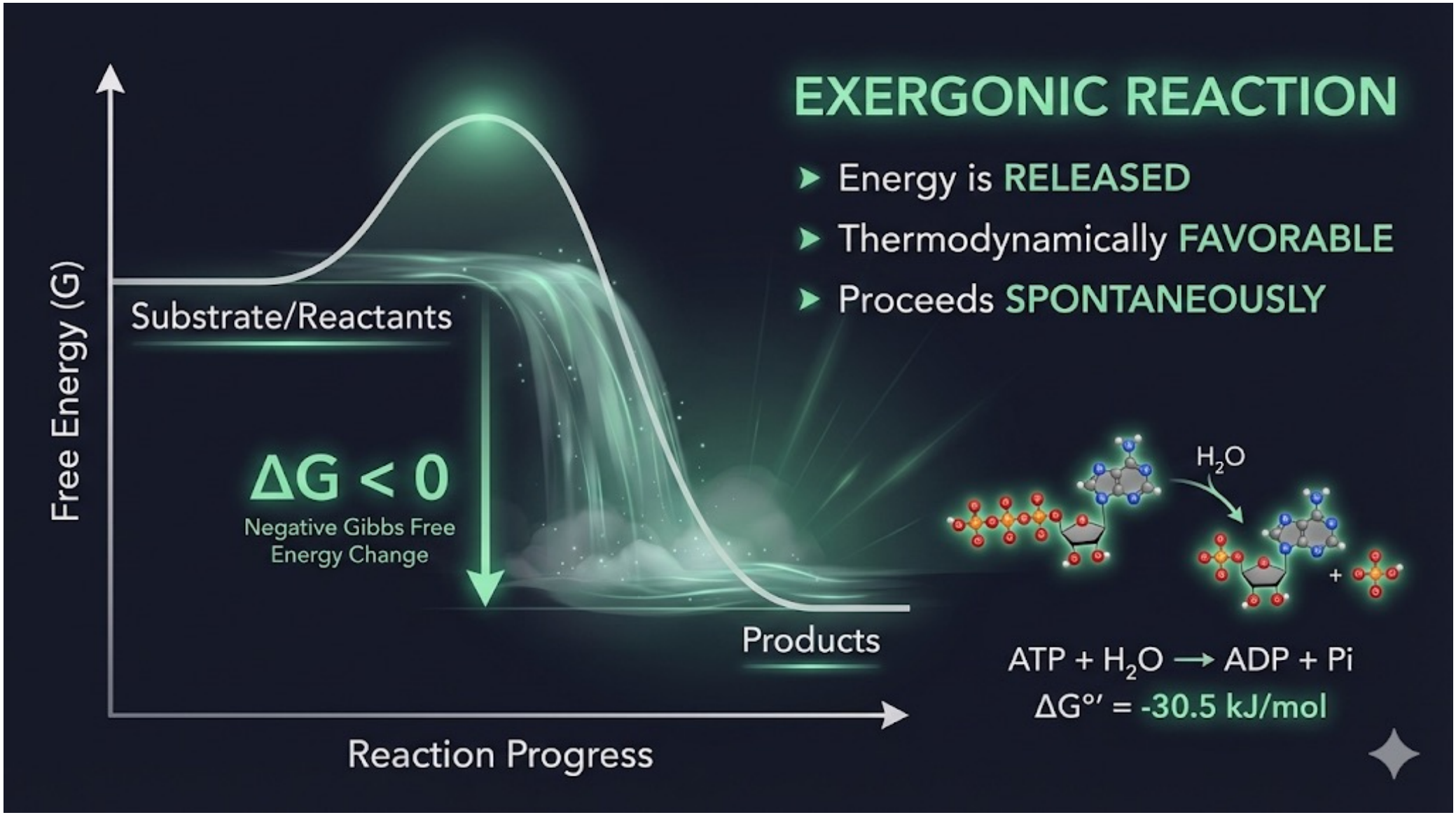


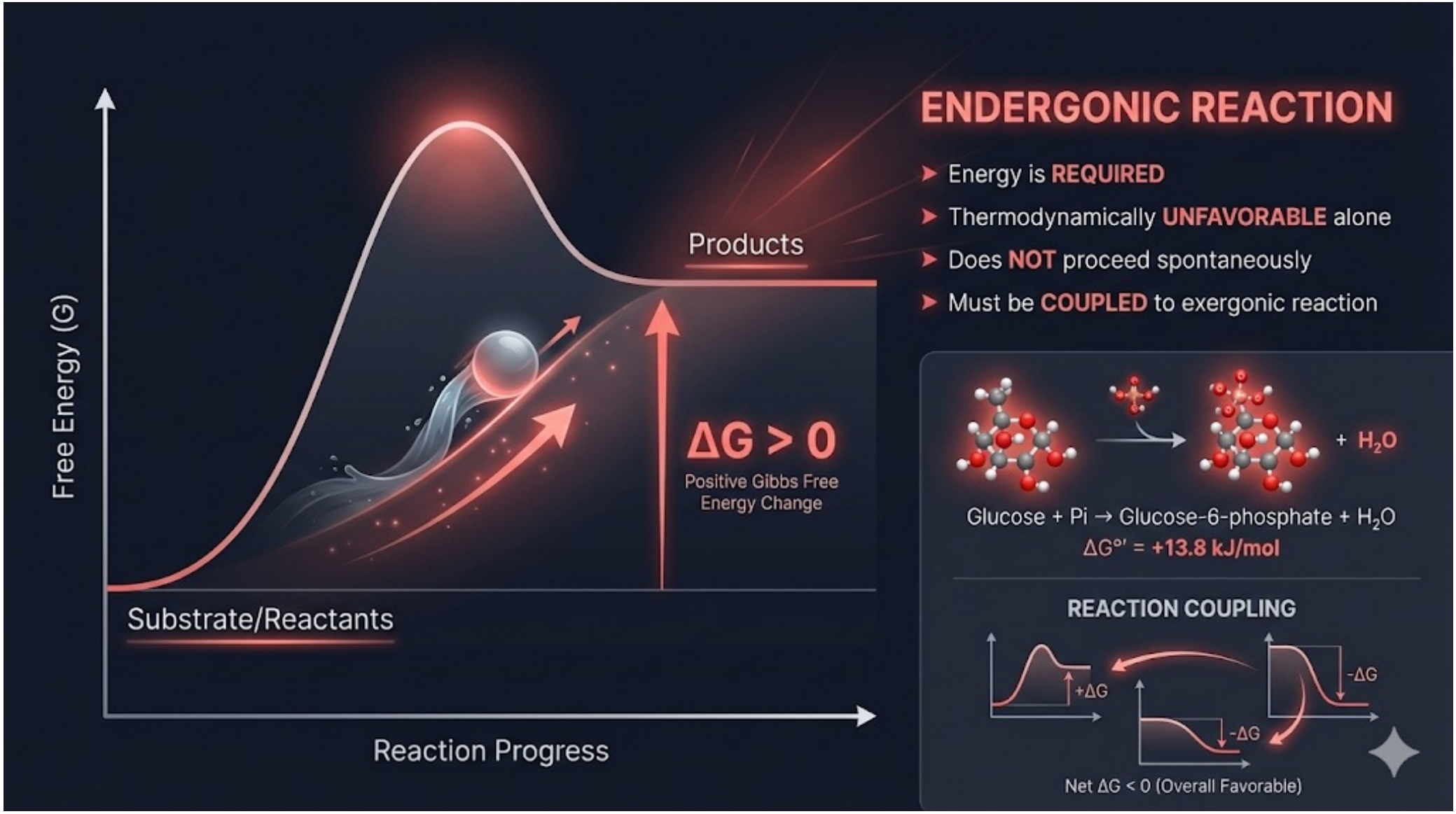
Enzymes cannot change the light color.
They only speed up the passage through a green light.

Reactions will spontaneously evolve toward something with lower energy
(2nd law of thermodynamics)



Spontaneous event involves a decrease in free energy, is exergonic
(thermodynamically favorable)


EXERGONIC REACTION
Energy is RELEASED
Thermodynamically FAVORABLE
Proceeds SPONTANEOUSLY
Free Energy (G)
Substrate/Reactants
ΔG < 0
Negative Gibbs Free Energy Change
Products
Reaction Progress
H2O
ATP + H2O ⟶ ADP + Pi
ΔG°′ = -30.5 kJ/mol


ENDERGONIC REACTION
Energy is REQUIRED
Thermodynamically UNFAVORABLE alone
Does NOT proceed spontaneously
Must be COUPLED to exergonic reaction
Free Energy (G)
Products
Substrate/Reactants
Reaction Progress
ΔG > 0
Positive Gibbs Free Energy Change
+ H2O
Glucose + Pi → Glucose-6-phosphate + H2O
ΔG°′ = +13.8 kJ/mol
REACTION COUPLING
+ΔG
-ΔG
-ΔG
Net ΔG < 0 (Overall Favorable)


MONEY IN THE BANK



(−) ΔH
exothermic



(+) ΔH
endothermic

Exo/endo THERMIC = heat. Exo/endo GONIC = free energy. Don't mix them up





The Free-Energy Change Provides Information About the Spontaneity but Not the Rate of a Reaction

- A reaction will occur without the input of energy, or spontaneously, only if ΔG is negative. Such reactions are called exergonic reactions.
- A reaction will not occur if the ΔG is positive. These reactions are called endergonic reactions.
- If a reaction is at equilibrium, there is no net change in the amount of reactant or product. At equilibrium, $\Delta G = 0$.
- The ΔG of a reaction depends only on the free energy difference between reactants and products and is independent of how the reaction occurs.
- The ΔG of a reaction provides no information about the rate of the reaction.

How do we tell if reaction is:

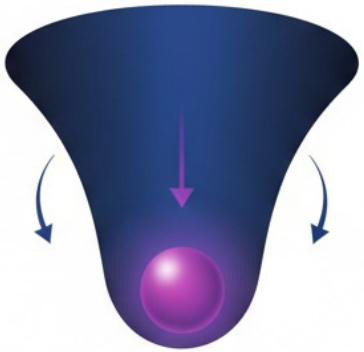
1. Spontaneous or not
2. Equilibrium constant
3. Directionality
4. Velocity

Equilibrium constant (K_{eq})

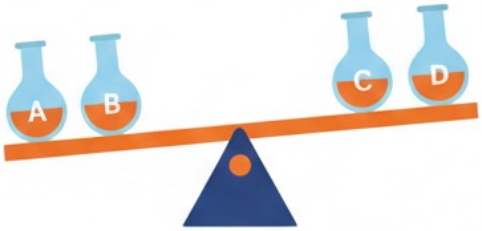
$$S \rightleftharpoons P$$

$$K_{eq} = \frac{[P]_{eq}}{[S]_{eq}}$$

Energy Perspective (ΔG)



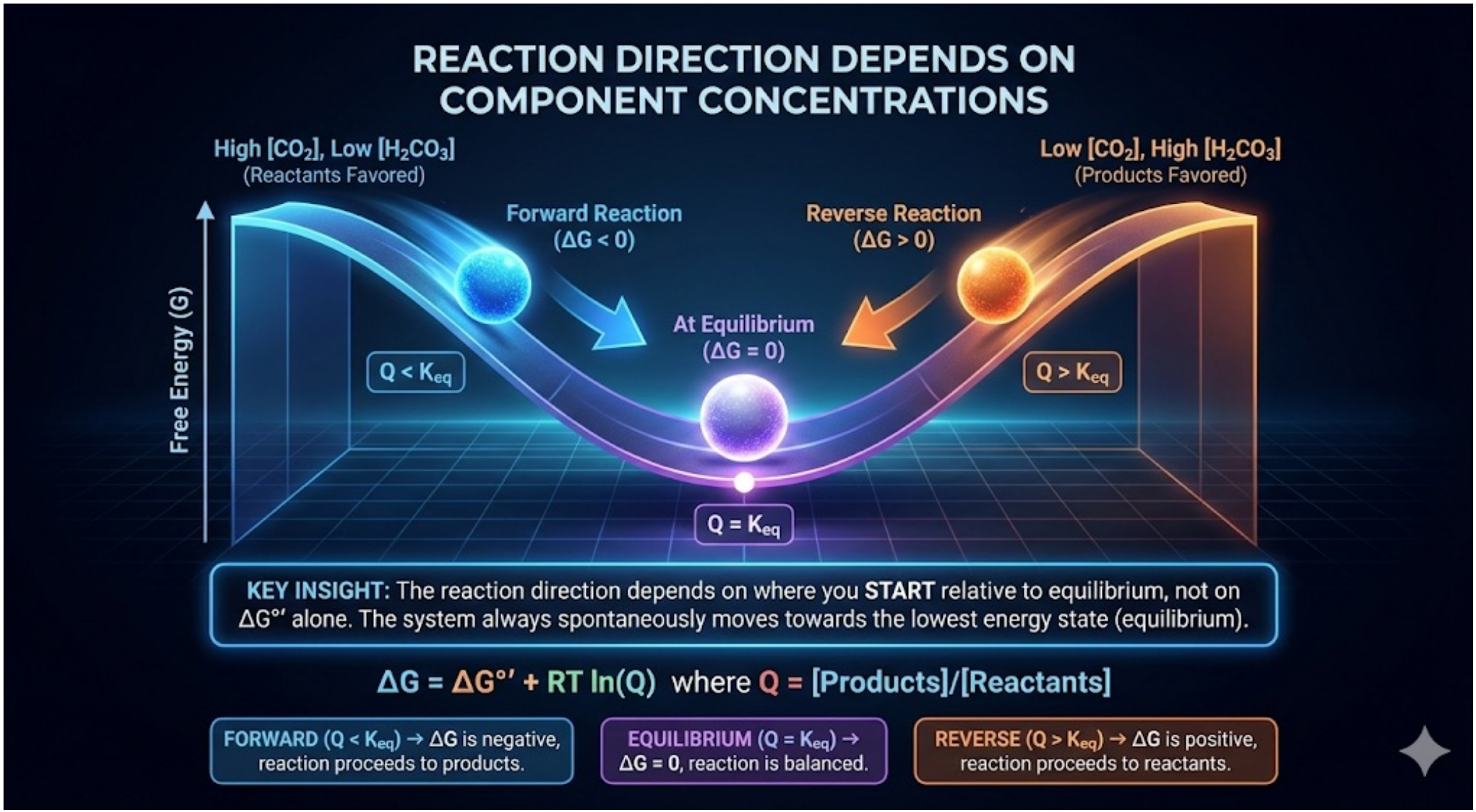
Concentation Perspective (Keq)



1. System “wants” to reach lowest energy state.
2. Lowest energy = Equiilblium.
3. Equilblium is a specific concentration ratio (Keq).

∴ Energy (ΔG) ⇌ Concentration Ratio (Keq)

Directionality depends on the concentration of the components



PROBLEM STATEMENT:

Hexokinase catalyzes glucose phosphorylation with $\Delta G^{\circ\prime} = -17$ kJ/mol at 298K. Calculate K_{eq}.

SOLUTION STEPS:

$$\Delta G^{\circ\prime} = -RT \ln K_{eq}$$

$$\rightarrow K_{eq} = e^{(-\Delta G^{\circ\prime}/RT)}$$

$$\rightarrow K_{eq} = e^{[-(-17\ \text{kJ/mol} \times 1000\ \text{J/kJ}) \div (8.314\ \text{J/(mol·K)} \times 298\ \text{K})]}$$

$$K_{eq} = e^{(17{,}000\ \text{J/mol} \div 2{,}478\ \text{J/mol})} = e^{6.86}$$

$$\rightarrow \boxed{K_{eq} \approx 950}$$

INTERPRETATION:

$K_{eq} \approx 10^3$ means products favored 1000:1 at equilibrium.

The Standard Free-Energy Change of a Reaction Is Related to the Equilibrium Constant

- The more exergonic a reaction is, the larger the equilibrium constant will be. The more endergonic a reaction is, the smaller the equilibrium constant will be.

- Note that the ΔG of the reaction can be larger than, smaller than, or equal to $\Delta G^{\circ\prime}$, depending on the concentrations of the reactants and products.

- A *kilojoule* (kJ) is equal to 1000 J.
- A *joule* (J) is the amount of energy needed to apply a 1-newton force over a distance of 1 meter.
- A *kilocalorie* (kcal) is equal to 1000 cal.
- A *calorie* (cal) is equivalent to the amount of heat required to raise the temperature of 1 gram of water from 14.5°C to 15.5°C.
- 1 kJ = 0.239 kcal

TABLE 6.3 Relation between $\Delta G^{\circ\prime}$ and K'_{eq} (at 25°C)

	$\Delta G^{\circ\prime}$	
K'_{eq}	kJ mol^{-1}	kcal mol^{-1}
10^{-5}	28.53	6.82
10^{-4}	22.84	5.46
10^{-3}	17.11	4.09
10^{-2}	11.42	2.73
10^{-1}	5.69	1.36
1	0	0
10	−5.69	−1.36
10^{2}	−11.42	−2.73
10^{3}	−17.11	−4.09
10^{4}	−22.84	−5.46
10^{5}	−28.53	−6.82

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Quick Quiz



QUICK QUIZ

Which of the following two reactions will take place spontaneously? What are the $\Delta G°'$ values for the reverse reactions?

$A \rightarrow B \quad \Delta G°' = -10 \text{ kJ mol}^{-1}$

$C \rightarrow D \quad \Delta G°' = +10 \text{ kJ mol}^{-1}$

WHAT ENZYMES DO

✓ CAN DO

- Accelerate reactions
- Lower activation energy ($\Delta G^{\ddagger}$)
- Speed up equilibrium attainment

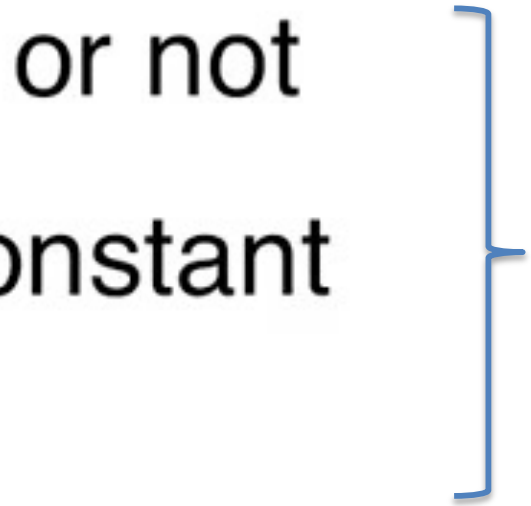


✗ CANNOT DO

- Change equilibrium position
- Change ΔG or $\Delta G^{\circ\prime}$
- Force unfavorable reactions

Same destination. Faster arrival.

1. Spontaneous or not
2. Equilibrium constant
3. Directionality
4. Velocity



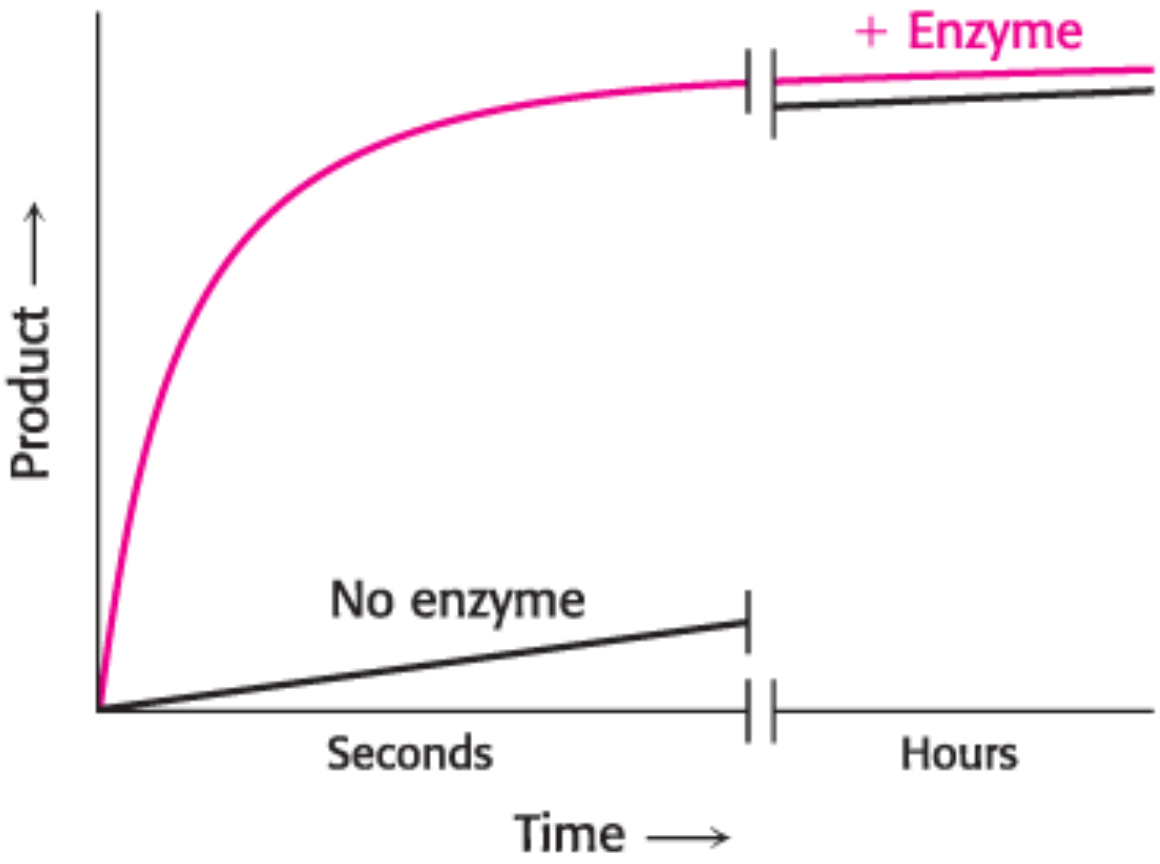
These qualities are about the rxn and are independent of the enzyme



This is where enzymes work

Enzymes Alter the Reaction Rate but Not the Reaction Equilibrium

- The reaction equilibrium is determined only by the free energy difference between the products and reactants. Enzymes cannot alter this difference.



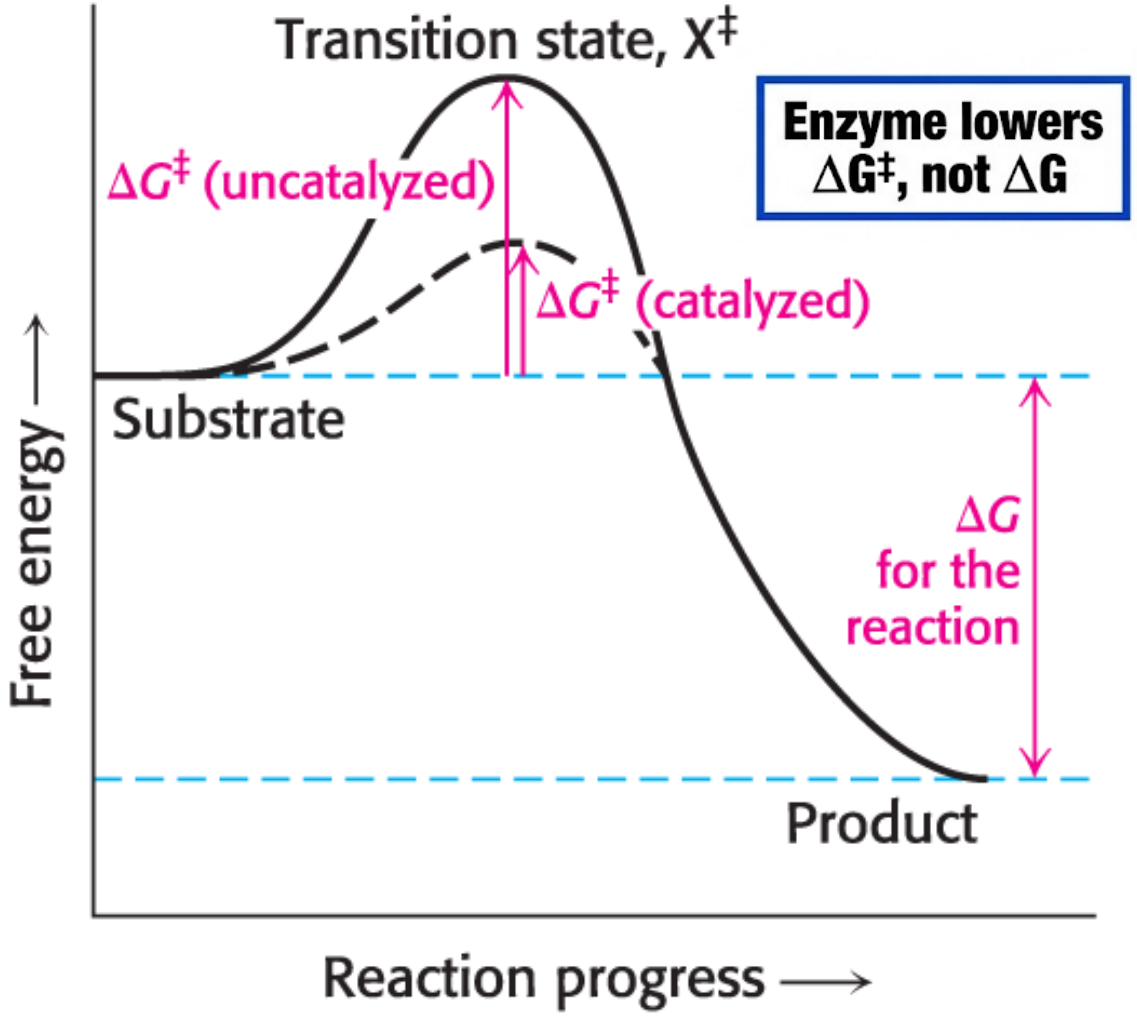

WHY do reactions happen?
Without enzyme—long way
With enzyme—shortcut
HOW do enzymes make them faster?
Coming up: The Transition State Secret

Section 7.4 Enzymes Facilitate the Formation of the Transition State

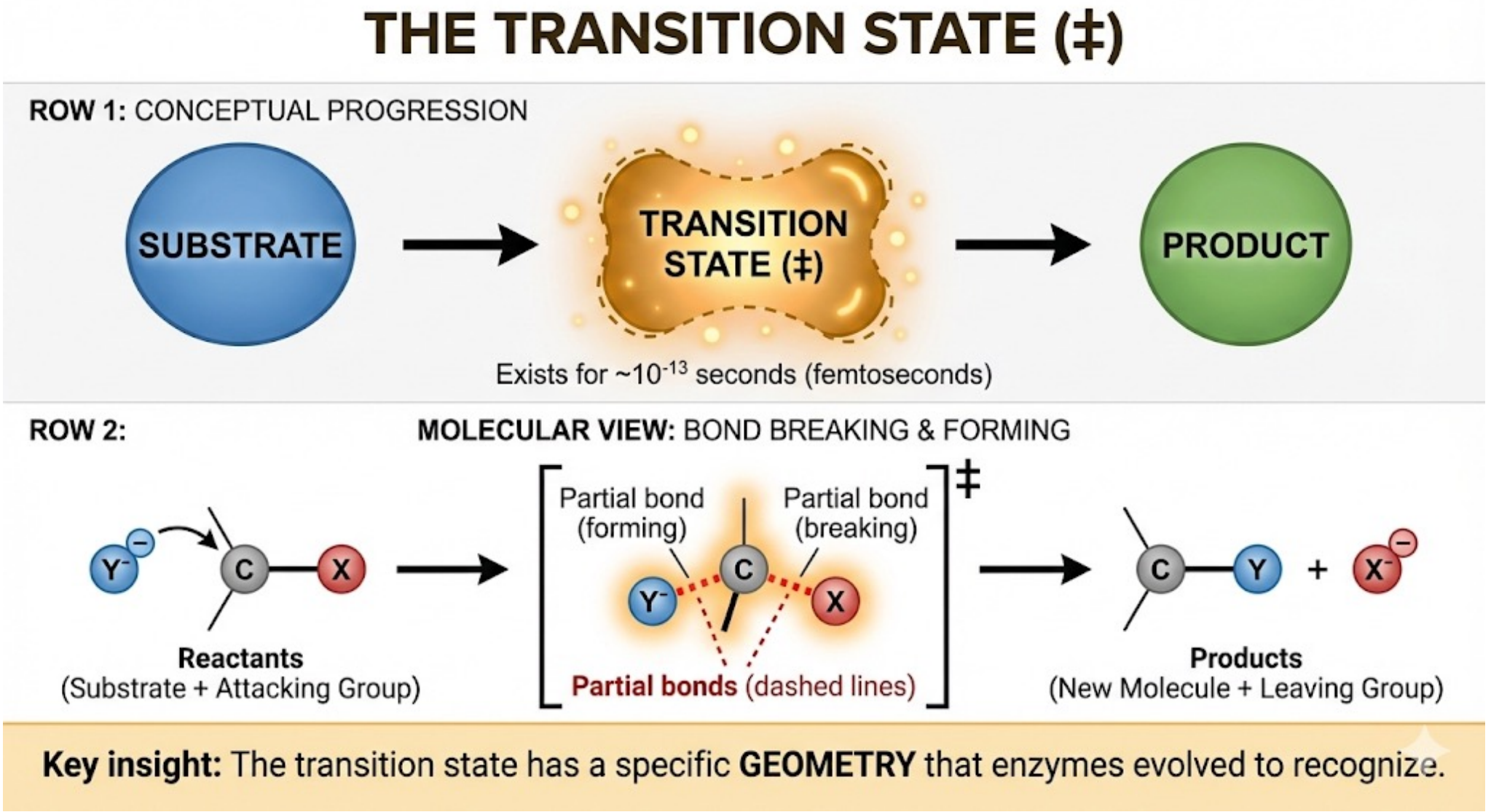
- **Learning objective 2: Explain the relation between the transition state and the active site of an enzyme and list the characteristics of active sites.**
- A chemical reaction proceeds through a transition state, a molecular form that is no longer substrate but not yet product.

- $S \rightleftharpoons X^{\ddagger} \rightarrow P$
  - The transition state is designated by the double dagger.
  - The energy required to form the transition state from the substrate is called the activation energy, symbolized by $\Delta G^{\ddagger}$.

- $\Delta G^{\ddagger} = G_X{}^{\ddagger} - G_S$
  - Enzymes facilitate the formation of the transition state.

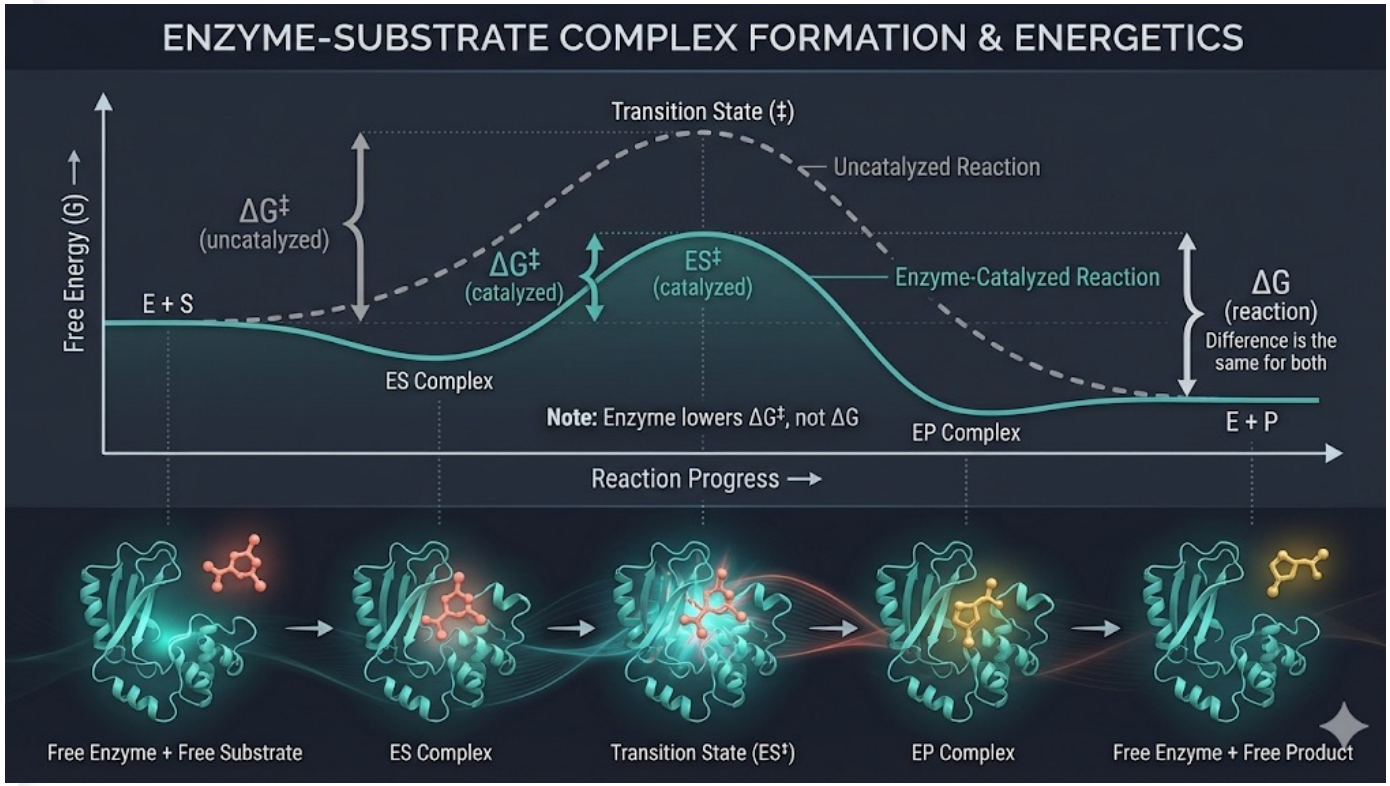


What is the transition state?



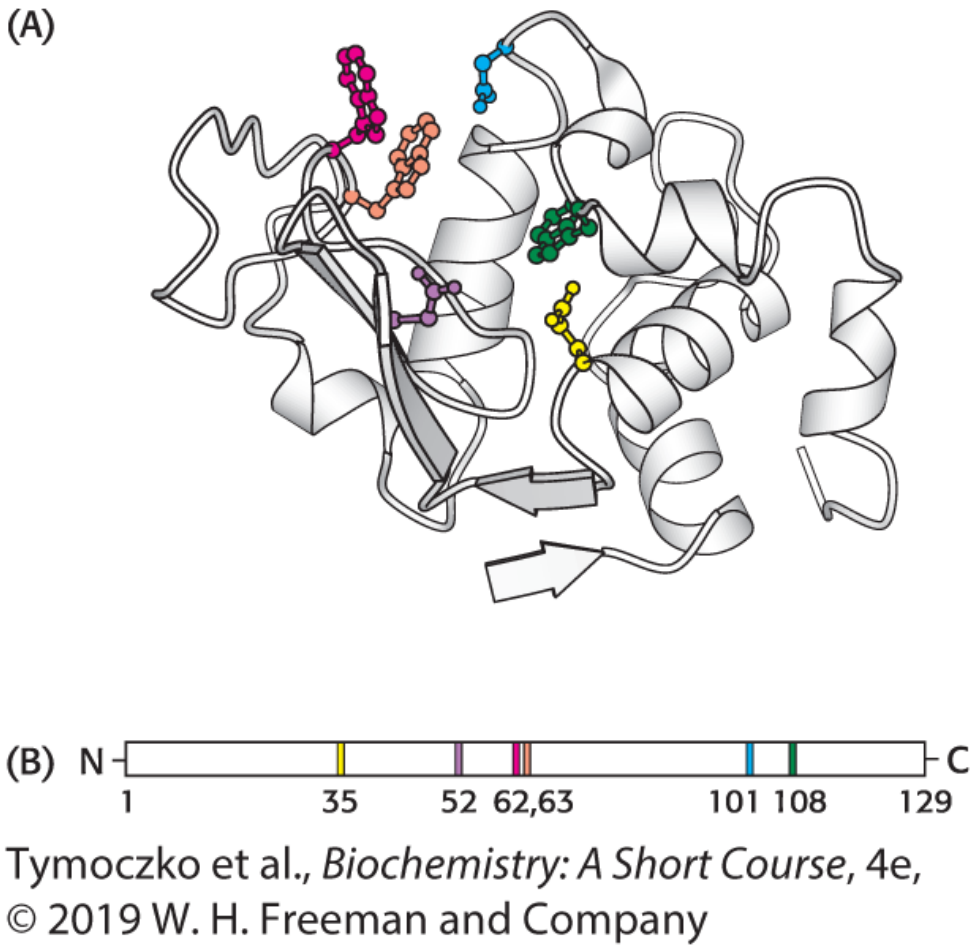
The Formation of an Enzyme–Substrate Complex Is the First Step in Enzymatic Catalysis

- Enzymes bring substrates together to form an enzyme–substrate complex on a particular region of the enzyme called the active site.
- The interaction of the enzyme and substrates at the active site promotes the formation of the transition state.



The Active Sites of Enzymes Have Some Common Features (1/2)

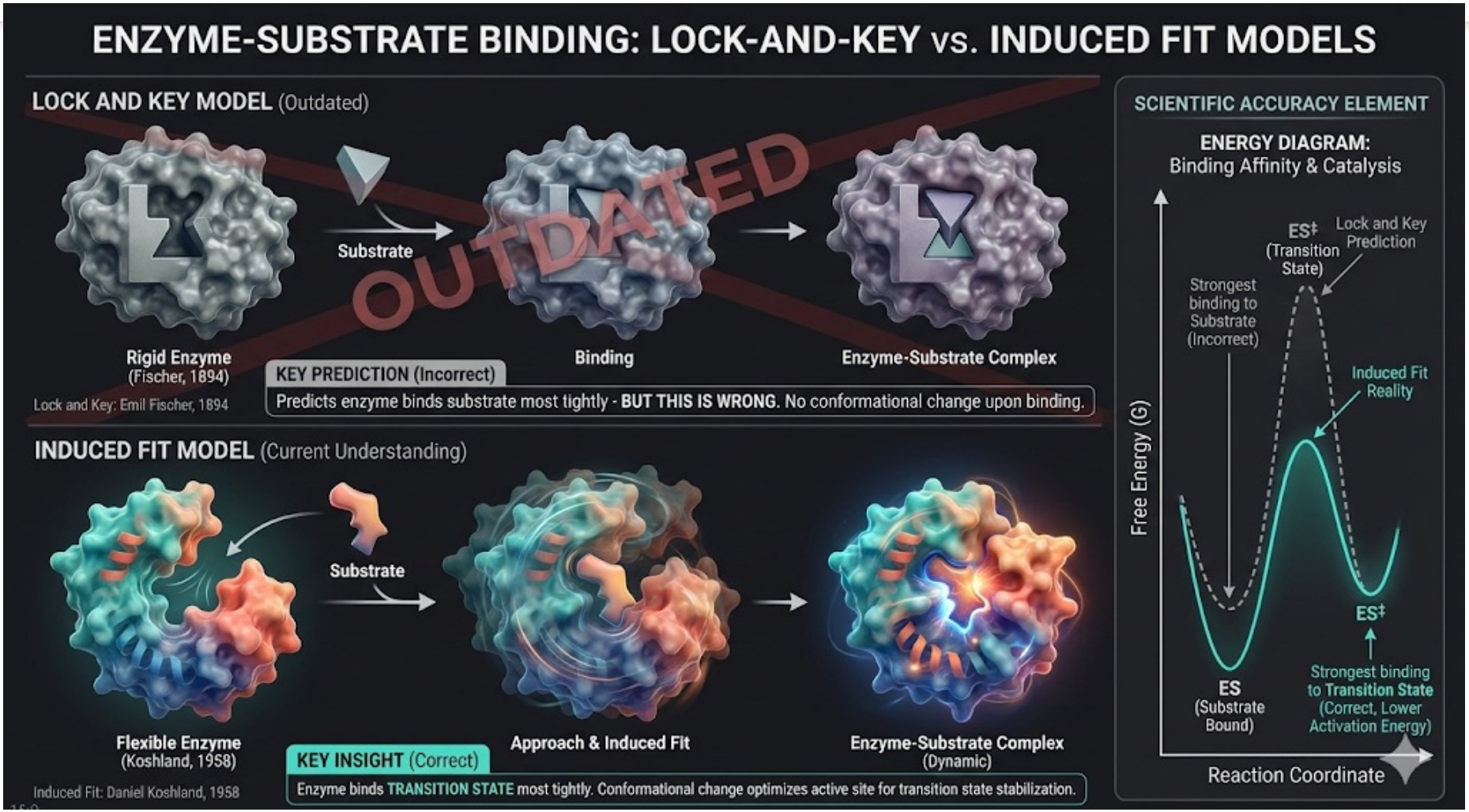
- The active site is a three-dimensional cleft or crevice created by amino acids from different parts of the primary structure.
- The active site constitutes a small portion of the enzyme volume.
- Active sites create unique microenvironments.
- The interaction of the enzyme and substrate at the active site involves multiple weak interactions.
- Enzyme specificity depends on the molecular architecture at the active site.



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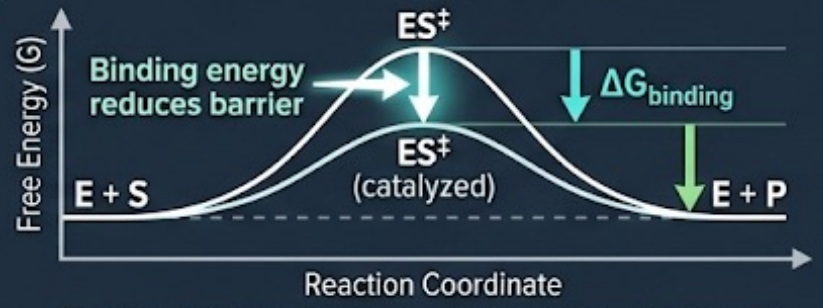
The Active Sites of Enzymes Have Some Common Features (2/2)

- Enzymes do not interact with their substrates like a lock and key.
- Rather, the enzyme changes shape upon substrate binding, a phenomenon called induced fit.



Binding energy facilitates catalysis

1 ENERGY LANDSCAPE



The binding energy between enzyme and transition state lowers the activation energy barrier.

2 SHAPE COMPLEMENTARITY



Substrate (#3498db)

Transition State (#f39c12)

Active site (#00cec9)

Active site is complementary to TRANSITION STATE, not substrate.

3 BINDING ENERGY EQUATION

$$\Delta G^{\ddagger}_{catalyzed} = \Delta G^{\ddagger}_{uncatalyzed} - \Delta G_{binding}$$



Tighter binding to TS → More energy released → Lower barrier

4 CLINICAL APPLICATION: Transition-State Analogs as Drugs



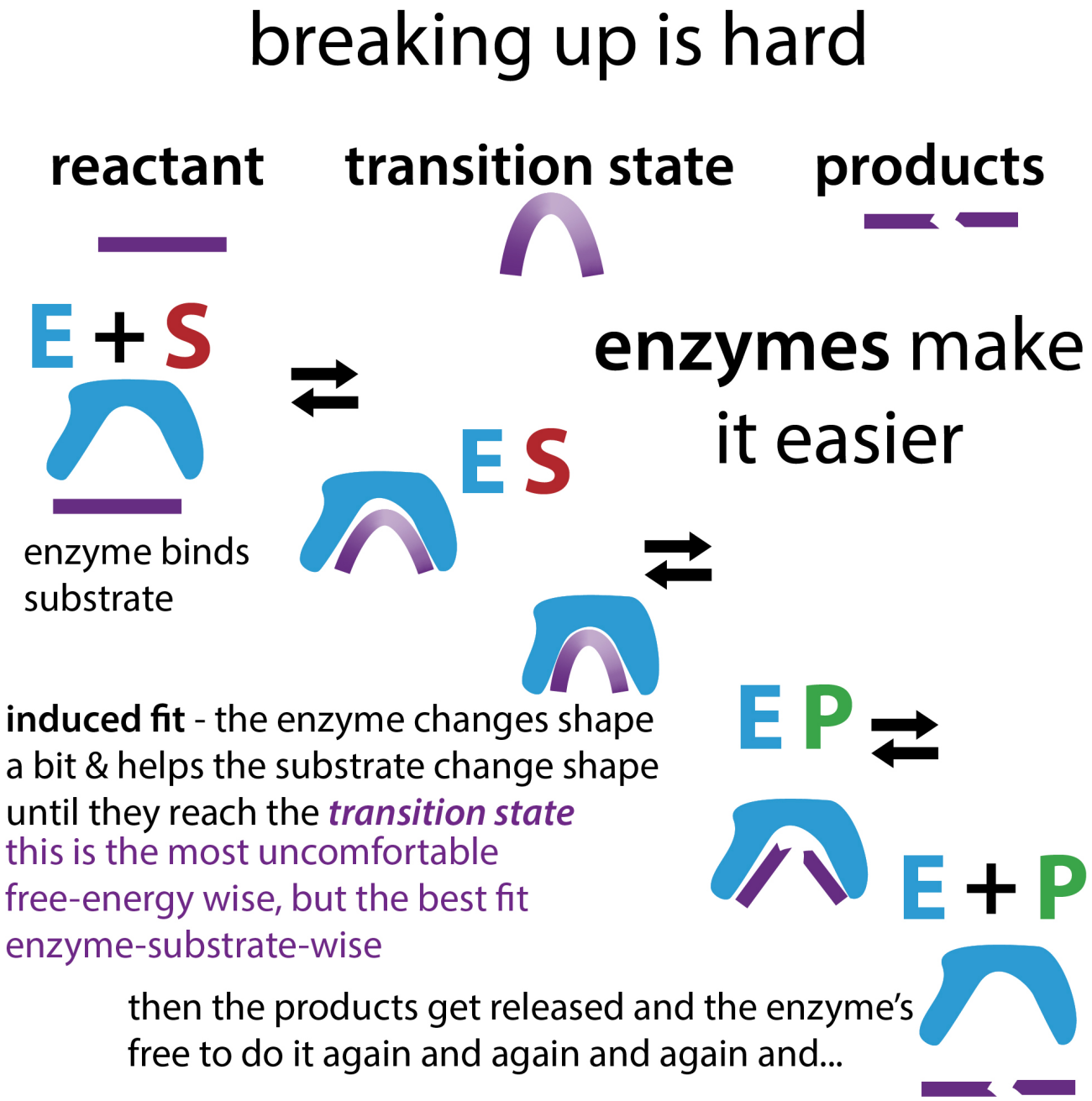
Drugs like HIV protease inhibitors mimic TS geometry → Bind 10^6× tighter than substrate → Potent inhibition

By mimicking the transition state, drugs can tightly bind and inhibit the enzyme.

The enzyme's job: Bind the transition state tighter than the substrate.

The Binding Energy Between Enzyme and Substrate Is Important for Catalysis

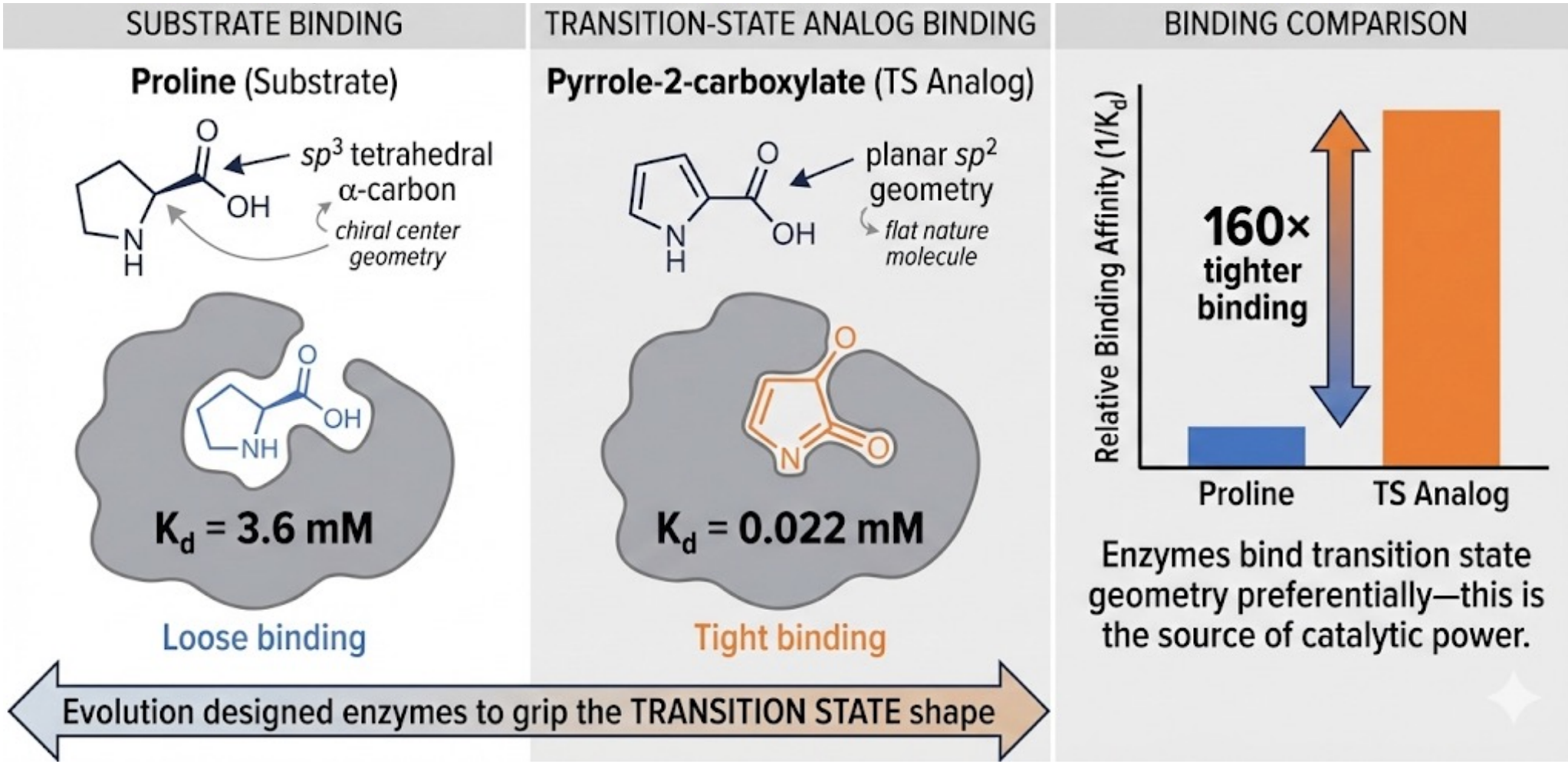
- Binding energy is the free energy released upon interaction of the enzyme and substrate.
- Binding energy is greatest when the enzyme interacts with the transition state, thus facilitating the formation of the transition state.
- Transition-State Analogs Are Potent Inhibitors of Enzymes



induced fit - the enzyme changes shape a bit & helps the substrate change shape until they reach the ***transition state***
this is the most uncomfortable free-energy wise, but the best fit enzyme-substrate-wise

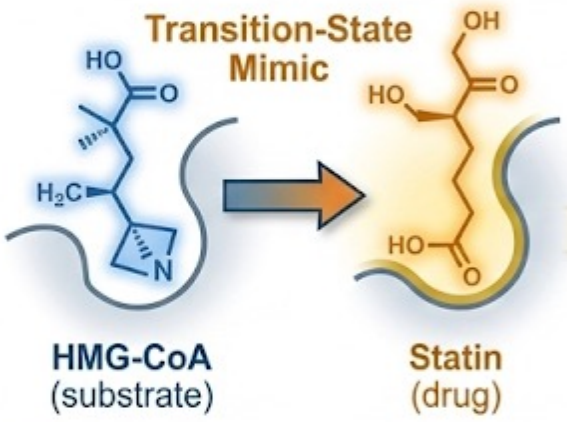
then the products get released and the enzyme's free to do it again and again and again and...

Transition state analog evidence.



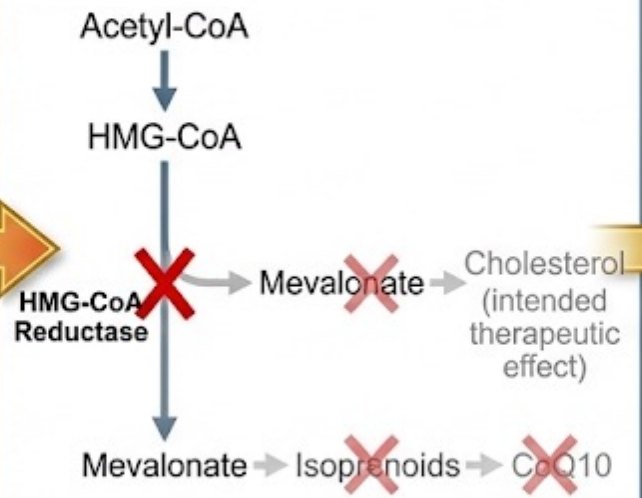
CLINICAL CASE RESOLUTION: The Statin Mystery Solved

1 1. STATINS ARE TRANSITION-STATE ANALOGS



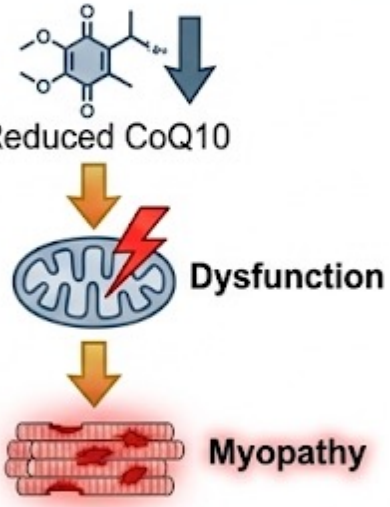
Statins bind enzyme active site.

2 2. BLOCKS CHOLESTEROL AND ISOPRENOID SYNTHESIS



Inhibition of both pathways.

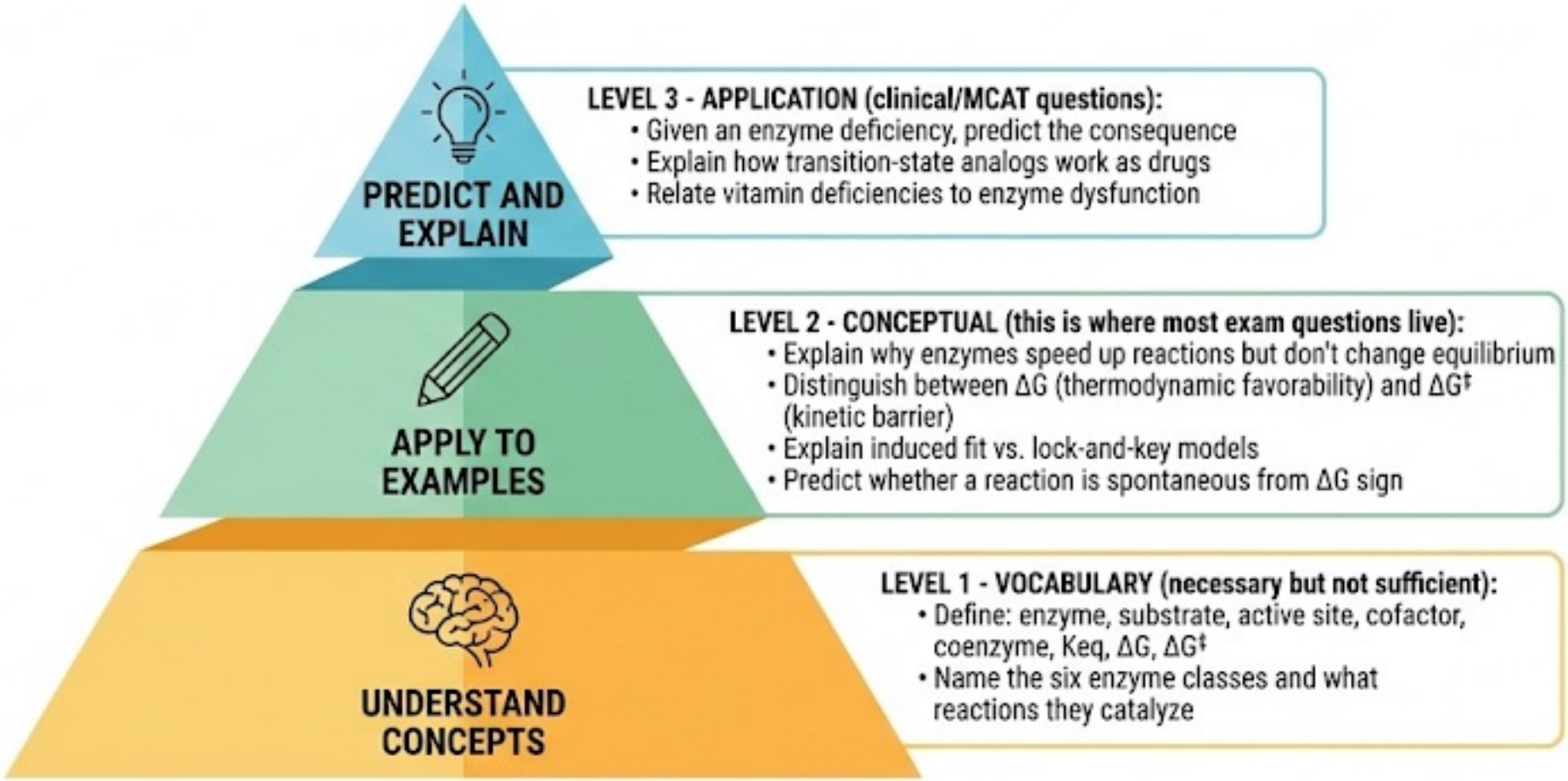
3 3. REDUCED CoQ10 → MITOCHONDRIAL DYSFUNCTION → MYOPATHY



Mitochondrial dysfunction & oxidative stress.

Clinical Pearl: Statins bind HMG-CoA reductase **~10,000× tighter** than substrate—same principle as proline racemase example.

HOW TO STUDY THIS CHAPTER:
ENZYME FUNDAMENTALS STUDY STRATEGY



LEVEL 3 - APPLICATION (clinical/MCAT questions):
- Given an enzyme deficiency, predict the consequence
- Explain how transition-state analogs work as drugs
- Relate vitamin deficiencies to enzyme dysfunction

LEVEL 2 - CONCEPTUAL (this is where most exam questions live):
- Explain why enzymes speed up reactions but don't change equilibrium
- Distinguish between ΔG (thermodynamic favorability) and $\Delta G^{\ddagger}$ (kinetic barrier)
- Explain induced fit vs. lock-and-key models
- Predict whether a reaction is spontaneous from ΔG sign

LEVEL 1 - VOCABULARY (necessary but not sufficient):
- Define: enzyme, substrate, active site, cofactor, coenzyme, Keq, ΔG, $\Delta G^{\ddagger}$
- Name the six enzyme classes and what reactions they catalyze

COMMON EXAM TRAPS
- Confusing rate with equilibrium
- Thinking negative ΔG means fast reaction (it doesn't—it means favorable)
- Believing enzymes "force" reactions in one direction
- Mixing up exergonic/endergonic with exothermic/endothermic